

COMPUTER SCIENCE XI

CHAPTER 1: COMPUTER SYSTEMS



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

یہ نوٹس خاص طور پر اساتذہ کے لیے تیار کیے گئے ہیں۔ حوالہ کے لیے نصابی کتاب کی پی ڈی ایف فائل "CS XI (1 & 2).pdf" کا استعمال کریں، جس کی من و عن پیروی کی گئی ہے۔ جن موضوعات پر ★ کا نشان ہے وہ امتحانات کے لیے انتہائی اہم ہیں۔

These notes are prepared specifically for teachers. For reference, please consult the textbook PDF file named exactly "**CS XI (1 & 2).pdf**", which has been followed verbatim. Topics marked with a ★ are highly critical for examinations.

1.1.1 Discrete and Continuous Quantities

Discrete Quantities: Separate, countable items with specific values. They can be counted one by one.

- Number of students in a classroom
- Number of cars in a parking lot

متفرد مقداریں (Discrete): یہ وہ مقداریں ہیں جنہیں گنا جا سکتا ہے اور ان کی مخصوص ویلیوز ہوتی ہیں۔

- کلاس میں طلباء کی تعداد
- پارکنگ میں گاڑیوں کی تعداد

Continuous Quantities: Can take any value within a range. They flow smoothly without breaks.

- Temperature of water
- Speed of a moving car

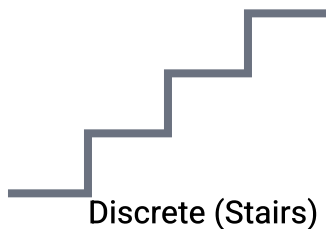
مسلسل مقداریں (Continuous): یہ کسی بھی حد کے اندر کوئی بھی ویلیو لے سکتی ہیں اور ان میں کوئی بریک نہیں ہوتا۔

- پانی کا درجہ حرارت
- چلتی ہوئی گاڑی کی رفتار

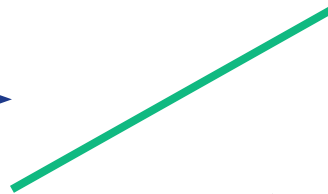
💡 Analogy: Stairs vs Ramp

Stairs represent discrete data (fixed steps). A ramp represents continuous data (any point).

سیڑھیاں متفرد (Discrete) کی مثال ہیں (مخصوص قدم)۔ ریمپ مسلسل (Continuous) کی مثال ہے (کسی بھی جگہ کھڑے ہو سکتے ہیں)۔



Discrete (Stairs)



Continuous (Ramp)

1.1.2 Introduction to Digital Systems

A digital system is an electronic system that uses only two values: **0** and **1**. These two values are called binary digits or bits.

ڈیجیٹل سسٹم ایک الیکٹرانک سسٹم ہے جو صرف دو ویلیوز 0 اور 1 کا استعمال کرتا ہے۔ ان ویلیوز کو بائنری ہندسے یا بٹس کہا جاتا ہے۔

- **0** represents OFF, LOW, or FALSE.
- **1** represents ON, HIGH, or TRUE.

Why Use Digital Systems? / ڈیجیٹل سسٹمز کا استعمال کیوں؟

Advantage	English Detail	Urdu Translation (اردو)
Reliable	Less affected by noise and interference.	شور اور مداخلت سے کم متاثر ہوتا ہے۔
Accurate	No loss of quality when copying data.	ڈیٹا کاپی کرتے وقت کو الٹی خراب نہیں ہوتی۔
Easy to Design	Only two states (0 and 1) to handle.	صرف دو حالتوں کو سنبھالنا آسان ہے۔
Programmable	Can be easily controlled by software.	سافٹ ویئر کے ذریعے آسانی سے کنٹرول کیا جا سکتا ہے۔

Examples of Digital Systems

- **Computers:** Processes data as 0s and 1s.
- **Smartphones:** Stores photos and music as binary data.
- **Digital Cameras:** Captures images as binary code.

1.1.3 Analog and Digital Signals

★ GOLDEN TOPIC

Analog Signals

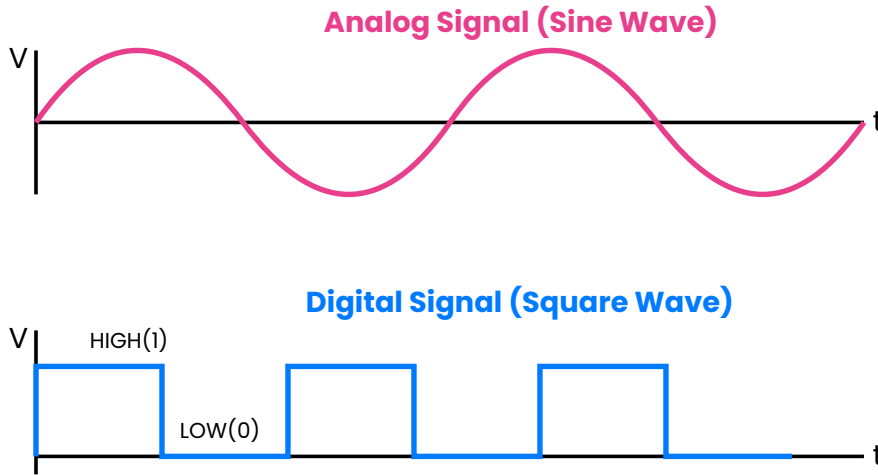
Continuous signals that change smoothly over time, like a sine wave. (e.g., Human voice, traditional thermometers).

مسلسل سگنلز جو وقت کے ساتھ ساتھ نرمی سے تبدیل ہوتے ہیں، جیسے لہر۔ (مثلاً انسانی آواز)۔

Digital Signals

Have only two levels: HIGH (1) and LOW (0). They jump instantly between values, creating a square wave. (e.g., Computer data).

ان میں صرف دو لیولز ہوتے ہیں: ہائی اور لو۔ یہ فوری تبدیل ہوتے ہیں، جس سے اسکوائر ویو بنتی ہے۔



Feature	Analog Data	Digital Data
Nature	Continuous flow.	Discrete, separate values (Bits).
Reliability	Low (affected by noise).	High (resistant to noise).

1.1.4 Boolean Algebra

Boolean algebra (developed by George Boole) deals with only two values: **TRUE (1)** and **FALSE (0)**. It is the foundation of digital electronics and computer programming.

بولین الجبرا (جارج بول کی ایجاد) میں صرف دو ویلیوز ہوتی ہیں: TRUE (1) اور FALSE (0)۔ یہ ڈیجیٹل الیکٹرانکس کی بنیاد ہے۔

Concept	Description	Example
Variables	Letters that hold values 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables & operations.	A AND B, $Y = A + B$

1.1.5 Boolean Operations

1. AND Operation (·)

The AND operation gives output 1 **ONLY when ALL inputs are 1**.

Expression: $Y = A \cdot B$

آؤٹ پٹ صرف اسی وقت 1 ہوگا جب تمام ان پٹس 1 ہوں۔

Real-life Example / روزمرہ کی مثال

A door with two locks. The door opens only when BOTH locks are unlocked.

ایک دروازہ جس پر دو تالے ہوں، دروازہ تب کھلے گا جب دونوں تالے کھلے ہوں۔

Boolean Operations (Continued)

2. OR Operation (+)

The OR operation gives output 1 **when ANY input is 1**.

Expression: $Y = A + B$

آؤٹ پٹ 1 ہوگا اگر کوئی ایک ان پٹ بھی 1 ہو۔

Real-life Example / روزمرہ کی مثال

A room with two doors. You can enter if EITHER door is open.

ایک کمرے کے دو دروازے ہیں، آپ کسی ایک سے بھی داخل ہو سکتے ہیں۔

3. NOT Operation (' or -)

The NOT operation **inverts the input**. It is a unary operation (operates on only one operand). If input is 1, output is 0.

Expression: $Y = A'$ or $Y = \bar{A}$

یہ ان پٹ کو الٹ دیتا ہے۔ اگر ان پٹ 1 ہے تو آؤٹ پٹ 0 ہوگا۔ یہ Unary آپریشن ہے۔

Real-life Example / روزمرہ کی مثال

A room light switch. When switch is ON, room is lit. When OFF, room is dark.

بجلی کا سوئچ۔ جب آن ہو تو بتی جلتی ہے، جب آف ہو تو بتی بجھ جاتی ہے۔

1.1.6 Truth Tables

A truth table shows all possible input combinations and their corresponding outputs for a logic gate or circuit.

ٹرٹھ ٹیبل کسی لاجک گیٹ یا سرکٹ کے تمام ممکنہ ان پٹس اور ان کے آؤٹ پٹس کو ظاہر کرتا ہے۔

How to create a truth table:

1. Count the number of inputs (n).
2. Calculate total rows = 2^n .
3. List all input combinations in binary order.
4. Calculate output for each combination.

2-Input Example (n=2, Rows=4)

A	B	Combinations
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

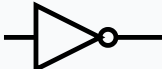
3-Input Example (n=3, Rows=8)

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1.1.7 Logic Gates (Basic Gates)

★ GOLDEN TOPIC

Logic gates are electronic circuits that perform Boolean operations. They are the building blocks of all digital systems.

Gate	Symbol	Expression	Truth Table															
AND آؤٹ پٹ 1 جب تمام ان پٹ 1 ہوں		$Y = A \cdot B$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Y	0	0	0	0	1	0	1	0	0	1	1	1
A	B	Y																
0	0	0																
0	1	0																
1	0	0																
1	1	1																
OR آؤٹ پٹ 1 اگر کوئی ایک ان پٹ 1 ہو		$Y = A + B$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Y	0	0	0	0	1	1	1	0	1	1	1	1
A	B	Y																
0	0	0																
0	1	1																
1	0	1																
1	1	1																
NOT ان پٹ کو الٹ دیتا ہے		$Y = A'$	<table><tr><th>A</th><th>Y</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	Y	0	1	1	0									
A	Y																	
0	1																	
1	0																	

Universal Logic Gates (NAND & NOR)

 **Teacher's Tip** / اساتذہ کے لیے مشورہ

Explain that NAND and NOR are called **Universal Gates** because ANY logic circuit in the world can be built using only these gates.

طلباء کو بتائیں کہ NAND اور NOR کو یونیورسل گیٹس کہا جاتا ہے کیونکہ دنیا کا کوئی بھی سرکٹ صرف ان کی مدد سے بنایا جاسکتا ہے۔

Gate	Symbol	Expression	Truth Table															
NAND کالٹ ہے AND یہ (NOT AND)		$Y = (A \cdot B)'$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Y	0	0	1	0	1	1	1	0	1	1	1	0
A	B	Y																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR کالٹ ہے OR یہ (NOT OR)		$Y = (A + B)'$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Y	0	0	1	0	1	0	1	0	0	1	1	0
A	B	Y																
0	0	1																
0	1	0																
1	0	0																
1	1	0																

Advanced Logic Gates (XOR & XNOR)

Gate	Symbol	Expression	Truth Table															
XOR آؤٹ پٹ 1 جب ان پٹس مختلف ہوں		$Y = A \oplus B$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	Y	0	0	0	0	1	1	1	0	1	1	1	0
A	B	Y																
0	0	0																
0	1	1																
1	0	1																
1	1	0																
XNOR آؤٹ پٹ 1 جب ان پٹس ایک جیسے ہوں		$Y = (A \oplus B)'$	<table><tr><th>A</th><th>B</th><th>Y</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	Y	0	0	1	0	1	0	1	0	0	1	1	1
A	B	Y																
0	0	1																
0	1	0																
1	0	0																
1	1	1																

Memory Trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the SAME.

1.1.8 & 1.1.10 Expressions & Minterms

Order of Precedence (Operations)

1. Parentheses (), 2. NOT ('), 3. AND ($\cdot$), 4. OR (+)

Minterms (SOP – Sum of Products)

A product (AND) of all variables that output a **1**. Complemented if value is 0 (A'), uncomplemented if 1 (A).

جہاں آؤٹ پٹ 1 ہو، وہاں Minterm استعمال ہوتا ہے۔ ان پٹ 0 ہو تو (A') لگاتے ہیں، 1 ہو تو ویسے ہی لکھتے ہیں۔

Maxterms (POS – Product of Sums)

A sum (OR) of all variables that output a **0**. Complemented if value is 1 (A'), uncomplemented if 0 (A).

جہاں آؤٹ پٹ 0 ہو، وہاں Maxterm استعمال ہوتا ہے۔

A	B	Output (Y)	Minterm (SOP)	Maxterm (POS)
0	0	0	–	$A + B$
0	1	1	$A'B$	–
1	0	0	–	$A' + B$
1	1	1	AB	–

Final SOP Expression: $Y = A'B + AB$

1.1.11 Logic Diagrams

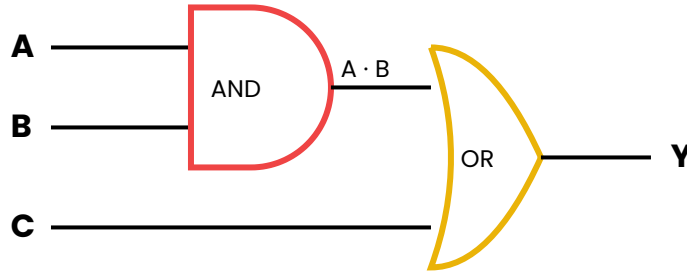
★ GOLDEN TOPIC

A logic diagram shows how logic gates are connected to create a circuit. Based on precedence, the operation with highest priority is drawn first.

لاجک ڈایا گرام یہ ظاہر کرتا ہے کہ لاجک گیٹس کو جوڑ کر ایک سرکٹ کیسے بنایا جاتا ہے۔ ترجیح (Precedence) کے لحاظ سے گیٹس بنائے جاتے ہیں۔

Example: Draw Logic Diagram for $Y = A \cdot B + C$

AND ($\cdot$) has higher precedence than OR ($+$). So, draw AND gate first.



1.1.13 Karnaugh Maps (K-Maps)

★ GOLDEN TOPIC

A K-map is a graphical tool used to simplify Boolean expressions. It uses a grid where adjacent cells differ by only ONE bit (Gray code). We group 1s in powers of 2 (1, 2, 4, 8) to simplify equations without complex algebra.

کارنوف میپ (K-Map) مساوات کو آسان بنانے کا گرافیکل ٹول ہے۔ اس میں متصل سیلز صرف ایک بٹ کے لحاظ سے مختلف ہوتے ہیں (گرے کوڈ)۔ ہم 1s کے گروپس بناتے ہیں۔

Two-Variable K-Map (Variables: A, B)

Needs $2^2 = 4$ cells.

	B = 0	B = 1
A = 0	Cell 0 (A'B')	Cell 1 (A'B)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Grouping Rule

Make as large groups as possible (of size 1, 2, 4, 8). Overlapping is allowed. Variables that change state (0 to 1) within a group are eliminated.

Three-Variable K-Map (A, B, C)

For 3 variables, we need $2^3 = 8$ cells. The columns follow Gray Code sequence: **00, 01, 11, 10**. This ensures adjacent cells differ by only one variable.

تین ویری ایبلز کے لیے ہمیں 8 سیلز درکار ہیں۔ کالز گرے کوڈ (00, 01, 11, 10) کی ترتیب میں ہوتے ہیں، بانٹری میں نہیں۔

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	$m_0 (A'B'C')$	$m_1 (A'B'C)$	$m_3 (A'BC)$	$m_2 (A'BC')$
A = 1	$m_4 (AB'C')$	$m_5 (AB'C)$	$m_7 (ABC)$	$m_6 (ABC')$

Teacher's Tip / اساتذہ کے لیے مشورہ

Emphasize to students that wrapping around the edges is allowed! A cell on the far left can be grouped with a cell on the far right.

طلباء کو واضح کریں کہ کناروں (Edges) کو بھی آپس میں ملا کر گروپ بنایا جاسکتا ہے۔ دائیں کونے کا سیل بائیں کونے کے سیل کے ساتھ گروپ ہو سکتا ہے۔

LogiSim Evolution: Simulation Tool

LogiSim Evolution is a graphical tool for designing and simulating logic circuits. It allows students to visually build the truth tables and boolean expressions they learn in theory.

1. The Interface Areas

- **Menu Bar:** Top bar for File, Edit, Project, Simulate.
- **Toolbar:** Quick access to standard tools (Wiring, Poke, AND, OR, NOT gates).
- **Explorer Panel (Left):** Contains component libraries (Gates, I/O, Wiring).
- **Attribute Table (Bottom Left):** Adjust properties of selected gates (e.g., Number of Inputs, Facing direction).
- **Drawing Canvas (Right):** The dotted grid where you place and wire gates.

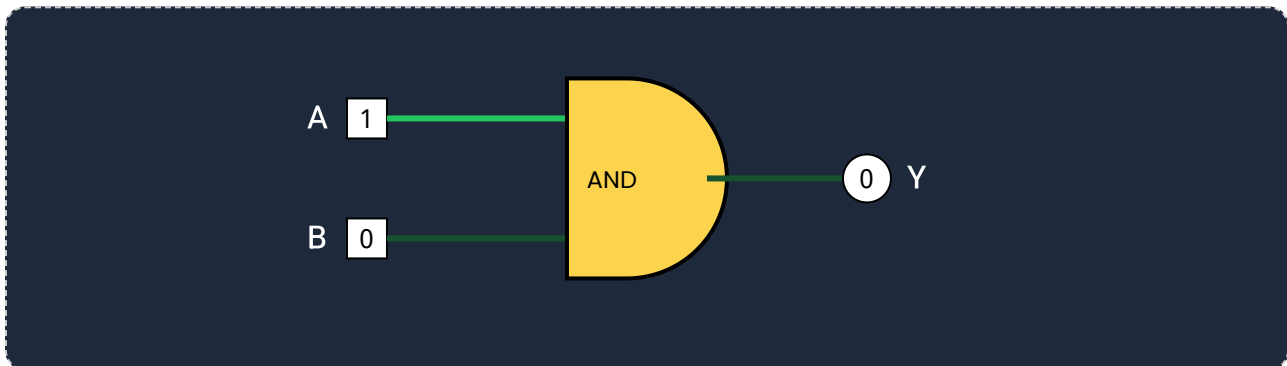
Wiring Colors during Simulation

- **Bright Green:** Logic 1 (HIGH)
- **Dark Green:** Logic 0 (LOW)
- **Blue:** Unknown / Unconnected pin
- **Red:** Error / Short circuit

Building a Circuit in LogiSim

Step-by-Step: Building $Y = A \cdot B$

1. Select the **AND Gate** from the Toolbar or Gates library and place it on the canvas.
2. Select the **Input Pin** (square icon with a 0) and place two pins to the left of the gate (label them A and B).
3. Select the **Output Pin** (circle icon with a 0) and place it to the right (label it Y).
4. Use the **Wiring Tool** to click and drag wires from the Input Pins to the AND gate's input ports. Wires must be horizontal or vertical.
5. Wire the gate's output to the Output Pin Y.
6. Select the **Poke Tool** (pointing finger) to toggle input A and B between 0 and 1 to test the circuit!



Circuit testing with Poke Tool. Notice the bright green line (Logic 1) from Input A.

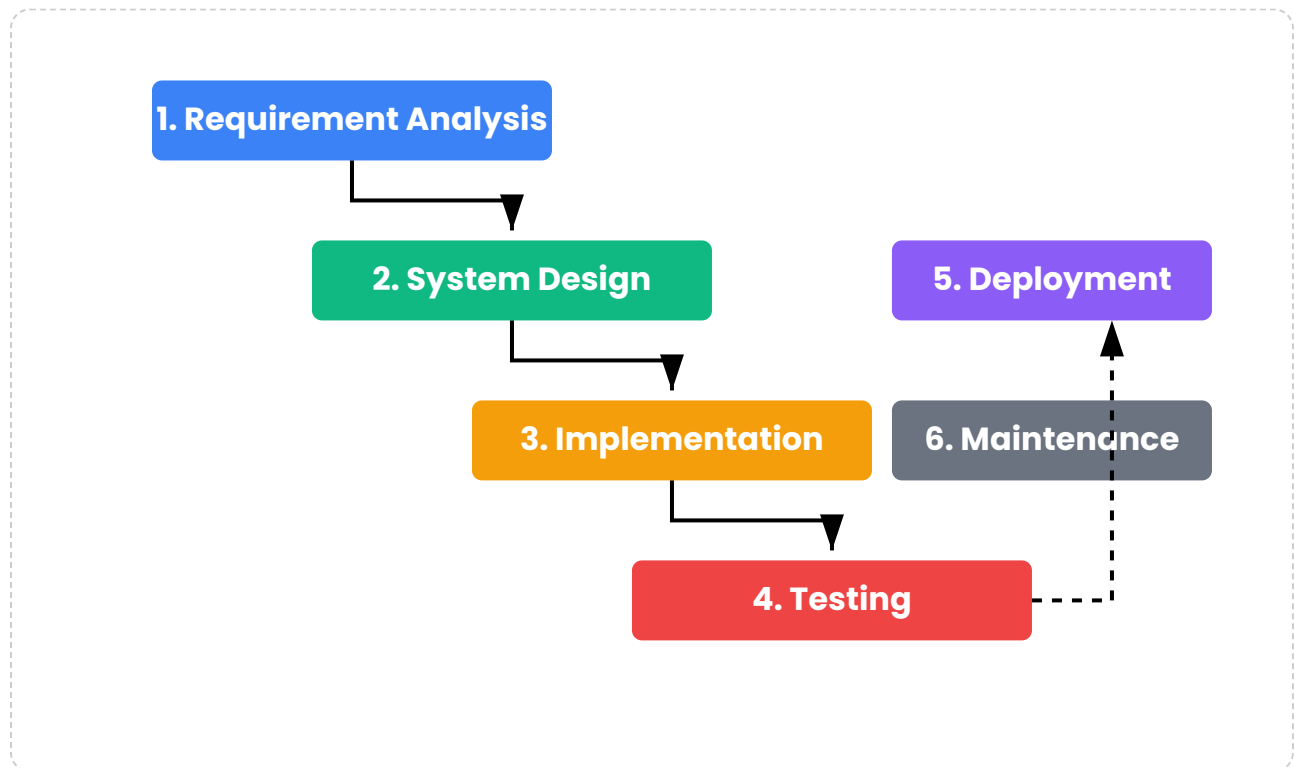
1.2 Software Development Life Cycle

★ GOLDEN TOPIC

SDLC is a structured process that guides software development from beginning to end. It ensures clear goals, better planning, risk reduction, and high quality.

ایس ڈی ایل سی (SDLC) ایک منظم عمل ہے جو سافٹ ویئر بنانے کی رہنمائی کرتا ہے۔ یہ واضح اہداف اور اعلیٰ معیار کو یقینی بناتا ہے۔

The 6 Phases of SDLC



Detailed SDLC Phases (1-3)

Phase	Activities & Details	Deliverable
1. Requirement Analysis منصوبہ بندی	Meet clients, understand needs, list features. This is the most important phase. صارف کی ضروریات کو سمجھنا اور فیچرز طے کرنا۔	SRS Document (Software Requirements Specification)
2. System Design ڈیزائن	Create High-Level Design (architecture), Low-Level Design (database, UI screens), Flowcharts. ڈیٹابیس اور اسکرین کا ڈیزائن بنانا۔	Design Document
3. Implementation کوڈنگ	Write actual code in programming languages (Python, Java, etc.) based on the design docs. پروگرامنگ لینگویج میں کوڈ لکھنا۔	Working Source Code



Teacher's Tip

Use the analogy of building a house: Phase 1 is asking the family what rooms they need. Phase 2 is the architect drawing blueprints. Phase 3 is the builders laying bricks.

Detailed SDLC Phases (4-6)

Phase	Activities & Details	Deliverable
4. Testing ٹیسٹنگ	Find and fix bugs. (See Black Box vs White Box below). کو تلاش کرنا اور ٹھیک کرنا۔ (Bugs) غلطیوں	Test Reports, Bug-free software
5. Deployment ڈپلائمنٹ	Release software to end-users (Direct, Phased, or Pilot release). سافٹ ویئر کو صارفین کے استعمال کے لیے پیش کرنا۔	Live Software
6. Maintenance دیکھ بھال	Provide support, updates, and fix post-release issues. سافٹ ویئر کی اپ ڈیٹس دینا اور مسائل حل کرنا۔	Patches, Updated Versions

Testing Methods



Black Box Testing

- Tester does NOT see code.
- Tests functionality (Inputs/Outputs).
- Done by dedicated testers.



White Box Testing

- Tester CAN see the code.
- Tests internal logic/structure.
- Done by developers.

1.2.4 The Waterfall Model

★ GOLDEN TOPIC

A linear, sequential approach. Each phase must be completed before the next begins. There is no going back to a previous phase easily.

یہ ایک سیدھا (Linear) ماڈل ہے۔ اگلا مرحلہ تب شروع ہوتا ہے جب پہلا مکمل ہو جائے۔ اس میں واپس جانا مشکل ہے۔

When to use?

- Requirements are clear, fixed, and well-understood.
- Project is small to medium size.
- Technology is stable.

Disadvantages (Cons)	Advantages (Pros)
Not flexible to changes.	Simple to understand and manage.
Testing happens very late in the process.	Well-documented at every step.

Case Study Reference: Library System

The textbook uses the "Al-Noor Library Management System" as a Waterfall case study because the requirements were 100% fixed before coding began.

1.2.4 The Agile Model

★ GOLDEN TOPIC

An iterative approach. Software is developed in small cycles called **sprints** (usually 2-4 weeks). Customer feedback is gathered continuously.

اس میں کام چھوٹے حصوں (Sprints) میں کیا جاتا ہے۔ یہ لچکدار (Flexible) ہوتا ہے اور تبدیلیاں آسانی سے کی جاسکتی ہیں۔

When to use?

- Requirements are unclear or may change.
- Customer wants frequent progress updates (early delivery).
- Complex or innovative projects.

Disadvantages (Cons)	Advantages (Pros)
Less documentation.	Highly flexible to changes.
Unpredictable end-time and cost.	Early delivery of working software.

Case Study Reference: Mobile App

The "Al-Noor Student Mobile Application" used Agile because teachers, parents, and students constantly requested new features during development.

Waterfall vs Agile Comparison

Dimension	Waterfall Model	Agile Model
Flexibility لچک	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer Involvement	Only at start (requirements) and end (delivery).	Continuous throughout the project (after every sprint).
Delivery of Software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

1.3 Communication Models (OSI)

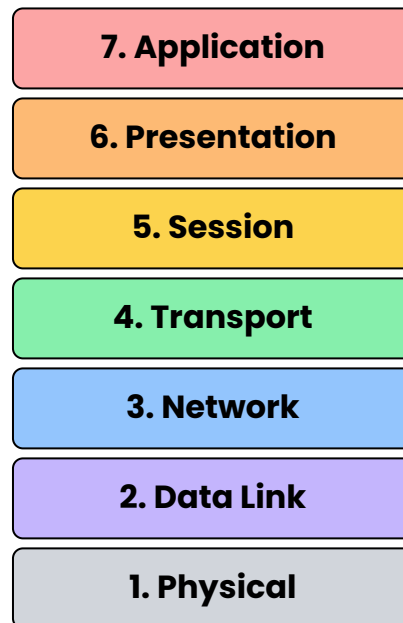
★ GOLDEN TOPIC

A communication model explains how data travels from one device to another through a network. The **OSI (Open Systems Interconnection) Model** was developed by ISO in 1984. It divides communication into 7 layers.

کیونیکیشن ماڈل بتاتا ہے کہ ایک ڈیوائس سے دوسری ڈیوائس تک ڈیٹا کیسے سفر کرتا ہے۔ او ایس آئی (OSI) ماڈل کی 7 تہیں (Layers) ہیں۔

🧠 Memory Trick / یاد رکھنے کا طریقہ

"Please Do Not Throw Sausage Pizza Away" (Bottom to Top: Physical, Data Link, Network, Transport, Session, Presentation, Application).



OSI Model Layers (Top Layers)

Layer	Main Job / بنیادی کام	Protocols
7. Application	Provides network services directly to user applications. صارف کی ایپس (ویب، ای میل) کو سروسز دینا۔	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data Translation, Encryption (security), and Compression. ڈیٹا کا ترجمہ، خفیہ کاری، اور سائز کم کرنا۔	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains, and closes communication sessions. رابطہ قائم کرنا اور ختم کرنا۔	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, Error checking, Port numbering. ڈیٹا کی قابل بھروسہ ترسیل اور غلطیوں کی جانچ۔	TCP (reliable), UDP (fast)

TCP vs UDP at Transport Layer

Use **TCP** when accuracy matters (Web browsing, Emails). Use **UDP** when speed matters more than perfection (Video streaming, online gaming).

OSI Bottom Layers & TCP/IP Model

Layer	Main Job / بنیادی کام	Protocols / Hardware
3. Network	Logical addressing (IP) and routing packets across networks. آئی پی ایڈریس لگانا اور روٹنگ کرنا۔	IP, ICMP (Ping), Routers
2. Data Link	Physical addressing (MAC) and error detection in frames. ایڈریس لگانا۔ (MAC) میک	Ethernet, MAC, Switches
1. Physical	Transmits raw bits (0,1) over cables/wireless mediums. تاروں کے ذریعے بٹس بھیجنا۔	Cables, Hubs, Wi-Fi radio

TCP/IP Model vs OSI Model

The TCP/IP model is the practical model used on the real Internet. It condenses the 7 OSI layers into just 4 layers.

TCP/IP Model (4 Layers)	Corresponds to OSI Layers
1. Application	Application, Presentation, Session (7,6,5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2,1)

COMPUTER SCIENCE XI

CHAPTER 2: COMPUTATIONAL THINKING & ALGORITHM



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

یہ نوٹس خاص طور پر اساتذہ کے لیے تیار کیے گئے ہیں۔ حوالہ کے لیے نصابی کتاب کی پی ڈی ایف فائل "CS XI (1 & 2).pdf" کا استعمال کریں، جس کی من و عن پیروی کی گئی ہے۔ جن موضوعات پر ★ کا نشان ہے وہ امتحانات کے لیے انتہائی اہم ہیں۔

These notes are prepared specifically for teachers. For reference, please consult the textbook PDF file named exactly "**CS XI (1 & 2).pdf**", which has been followed verbatim. Topics marked with a ★ are highly critical for examinations.

2.1 Computational Thinking

Computational Thinking (CT) is a systematic approach to solving problems. It involves breaking problems down, identifying patterns, abstracting details, and developing algorithms.

کمپیوٹیشنل تھنکنگ (CT) مسائل کو حل کرنے کا ایک منظم طریقہ ہے۔ اس میں مسائل کو حصوں میں تقسیم کرنا، پیٹرن تلاش کرنا، غیر ضروری تفصیلات کو ہٹانا، اور الگورتھم بنانا شامل ہے۔

💡 Did you know?

Pseudocode is a human-readable representation of an algorithm using plain English and structured programming keywords. It's the bridge between human logic and machine code.

2.2 Algorithm

★ GOLDEN TOPIC

An **algorithm** is a step-by-step logical procedure used to solve a problem. It is conceptual in nature and completely independent of any programming language.

الگورتھم کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ کار ہے۔ یہ تصوراتی ہوتا ہے اور کسی بھی پروگرامنگ لینگویج سے مکمل آزاد ہوتا ہے۔

2.2.1 Importance of Algorithms

- Provide a structured approach to problem-solving, making complex tasks manageable.
- Optimize time and memory usage, crucial for large-scale applications.
- Can be reused for similar problems, saving time and effort.
- Ensure accuracy and consistency, reducing human error.
- Act as a blueprint before actual code implementation.

2.2.2 Algorithm vs Pseudocode

★ GOLDEN TOPIC

While both are used in problem-solving, they serve different purposes in the design phase.

اگرچہ دونوں مسائل حل کرنے میں استعمال ہوتے ہیں، لیکن ڈیزائننگ کے مرحلے میں ان کے مقاصد مختلف ہوتے ہیں۔

Algorithm (الگورتھم)	Pseudocode (سوڈو کوڈ)
A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Conceptual and language-independent.	Semi-formal and structured but not executable code.
To define what needs to be done logically.	To present how it can be implemented in a readable format.
Usually described in plain steps, bullets, or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Teacher's Explanation / اساتذہ کے لیے وضاحت

Use this analogy: An **Algorithm** is like a recipe (e.g., "Bake a cake at 350 degrees"). **Pseudocode** is like a structured kitchen manual using standard verbs (e.g., SET temperature TO 350; WHILE cake is not baked, WAIT).

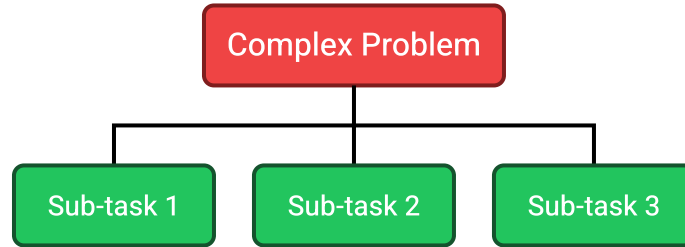
2.3 Algorithmic Strategies for Problem Solving

2.3.1 Decomposition

★ GOLDEN TOPIC

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts. By dividing a big problem into simpler steps, it becomes easier to understand, develop, and solve.

ڈی کمپوزیشن (Decomposition) کسی پیچیدہ مسئلے یا ٹاسک کو چھوٹے اور قابل انتظام حصوں میں تقسیم کرنے کا نام ہے۔ اس سے مسئلے کو سمجھنا اور حل کرنا آسان ہو جاتا ہے۔



Real-Life Example: Learning a New Language

Subtask	Description
Learn Vocabulary	Learn character sets for people, actions, things (e.g., "I", "eat", "apple").
Grammar Rules	Learn structure: Subject → Verb → Object.
Learn Tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence Building	Combine words using rules to create valid sentences.

Why Decomposition is Useful?

- **Reduces complexity:** Focus on one piece at a time.
- **Helps in debugging:** Find exactly where the mistake is (e.g., verb tense or word order).
- **Promotes reusability:** Once you learn a small part, you can apply it elsewhere.

2.3.2 Pattern Recognition

Pattern Recognition is the process of identifying regularities, repetitions, or rules in data. It helps in reducing overall algorithm complexity and forms the foundation of concepts such as loops.

پیٹرن ریکگنیشن ڈیٹا میں تکرار (Repetition) یا اصولوں کو پہچاننے کا عمل ہے۔ یہ الگورتھم کی پیچیدگی کو کم کرتا ہے اور لوپس (Loops) جیسی چیزوں کی بنیاد ہے۔

How it works? Ask yourself:

- What is repeating?
- What changes in a predictable way?
- Can I generalize this?

Example: Triangle of Stars

```
  *
 * *
* * *
* * * *
* * * * *
```

In this example, the pattern shows that each row adds 1 star to the previous row's count.

- Row 1 = 1 star
- Row 2 = 2 stars
- Row N = N stars

Recognizing this pattern allows us to write a generic loop algorithm instead of writing 5 separate print statements.

2.3.3 Abstraction

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem. It reduces complexity by ignoring what is not important (noise).

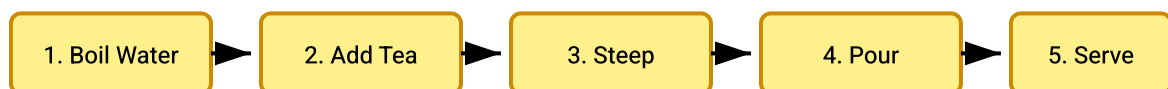
ایبسٹرکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپانا اور صرف ان اہم باتوں پر توجہ دینا جو مسئلہ حل کرنے کے لیے ضروری ہوں۔

Example: Applying Abstraction to "Making Tea"

To make an algorithm for tea, we must separate the necessary steps from the situational changes and the noise.

1. Necessary Steps (Core Logic)	2. Changes based on situation (Variables)	3. Abstraction Items (Ignore/Noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few mins 4. Pour into cup 5. Serve	- Ingredients (Black, Green, Milk) - Additions (Sugar, Lemon) - Tools (Stove vs Kettle)	- Brand of kettle - Cup color or design - Where tea is made (Kitchen, Office)

The Core Abstracted Logic (Independent of cup color or kettle brand)



2.4 Sorting and Searching Algorithms

Both are fundamental techniques of computer science. They work on lists of the same data type.

یہ دونوں کمپیوٹر سائنس کی بنیادی تکنیکیں ہیں۔ یہ ایک ہی قسم کے ڈیٹا (جیسے نام یا عمر) کی فہرستوں پر کام کرتی ہیں۔

Sorting (ترتیب دینا)

Arranging a collection of items. May be Ascending (0-9, A-Z) or Descending (9-0, Z-A). Used to organize data.

Searching (تلاش کرنا)

Locating a specific element within a list to find its position or existence.

2.4.1 Bubble Sort

★ GOLDEN TOPIC

An elementary sorting algorithm that works by repeatedly comparing adjacent elements and swapping them if they are in the wrong order.

بل سورٹ ایک بنیادی الگورتھم ہے جو ساتھ والے (Adjacent) ایلیمنٹس کا موازنہ کرتا ہے اور اگر وہ غلط ترتیب میں ہوں تو انہیں آپس میں بدل (Swap) دیتا ہے۔

Algorithm

```
Step 1: Start
Step 2: Set DATA := {8,4,1,9,3}
Step 3: Set N := 5
Step 4: Repeat Step 5 and 6 for
K=1 to N-1
Step 5: Set PTR := 1
Step 6: Repeat while PTR ≤ N-K:
  a: If DATA[PTR] > DATA[PTR+1]:
    1. Set temp = DATA[PTR]
    2. Set DATA[PTR] =
DATA[PTR+1]
    3. Set DATA[PTR+1] = temp
  b: Set PTR := PTR + 1
Step 7: End
```

Pseudocode

```
Start
DATA = {8,4,1,9,3}
N = 5
For outer = 0 to N-1
  For inner = 0 to N-2-outer
    IF DATA[inner] > DATA[inner+1]
      Temp = DATA[inner]
      DATA[inner] = DATA[inner+1]
      DATA[inner+1] = Temp
    End If
  End For
End For
End
```

Bubble Sort Example

Unsorted List: 8, 4, 1, 9, 3

First Step (Pass)	
[8, 4], 1, 9, 3	1st (8) > 2nd (4) → Swap → 4, 8, 1, 9, 3
4, [8, 1], 9, 3	2nd (8) > 3rd (1) → Swap → 4, 1, 8, 9, 3
4, 1, [8, 9], 3	3rd (8) NOT > 4th (9) → No Change
4, 1, 8, [9, 3]	4th (9) > 5th (3) → Swap → 4, 1, 8, 3, 9 (9 is now at its correct place)

Second Step (Pass)	
[4, 1], 8, 3, 9	1st (4) > 2nd (1) → Swap → 1, 4, 8, 3, 9
1, [4, 8], 3, 9	2nd (4) NOT > 3rd (8) → No Change
1, 4, [8, 3], 9	3rd (8) > 4th (3) → Swap → 1, 4, 3, 8, 9

Third & Fourth Step (Passes)	
1, [4, 3], 8, 9	(Step 3): 2nd (4) > 3rd (3) → Swap → 1, 3, 4, 8, 9
[1, 3], 4, 8, 9	(Step 4): 1st (1) NOT > 2nd (3) → No Change . List is sorted!

Sorted List: 1, 3, 4, 8, 9

Advantages

- Easiest algorithm to visualize and explain.
- Does not need extra memory (in-place sorting).
- Stable (equal items stay in original order).

Disadvantages

- Uses nested loops; time grows exponentially.
- Performs a massive amount of swaps.
- Useless for real-world "Big Data."

2.4.1 Selection Sort

★ GOLDEN TOPIC

A simple sorting algorithm that works by repeatedly selecting the *smallest* element from the unsorted portion of a list and moving it to its correct position at the front.

سیلیکشن سورت ایک سادہ الگورتھم ہے جو فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹے ایلیمنٹ کو تلاش کرتا ہے اور اسے شروع میں درست جگہ پر رکھ دیتا ہے۔

Algorithm

```
Step 1: Start
Step 2: Set DATA := {8,4,1,9,3}
Step 3: Set N := 5
Step 4: Repeat Step 5 and 6 for
K=1 to N
Step 5: Set PTR := K+1
Step 6: Repeat while PTR ≤ N:
  a: If DATA[K] > DATA[PTR]:
    1. Set temp = DATA[K]
    2. Set DATA[K] = DATA[PTR]
    3. Set DATA[PTR] = temp
  b: Set PTR := PTR + 1
Step 7: End
```

Pseudocode

```
Start
DATA = {8,4,1,9,3}
N = 5
For outer = 0 to N-1
  For inner = outer+1 to N-1
    IF DATA[outer] > DATA[inner]
      Temp = DATA[outer]
      DATA[outer] = DATA[inner]
      DATA[inner] = Temp
    End If
  End For
End For
End
```

Teacher's Tip

Point out the difference in the inner loop starting position compared to Bubble Sort. Selection sort inner loop starts at outer+1, whereas Bubble Sort always starts at 0.

Selection Sort Example

Unsorted List: 8, 4, 1, 9, 3

First Step (Finding smallest for Pos 1)	
[8], [4], 1, 9, 3	Pos1 (8) > Pos2 (4) → Swap → 4, 8, 1, 9, 3
[4], 8, [1], 9, 3	Pos1 (4) > Pos3 (1) → Swap → 1, 8, 4, 9, 3
[1], 8, 4, [9], 3	Pos1 (1) NOT > Pos4 (9) → No Change
[1], 8, 4, 9, [3]	Pos1 (1) NOT > Pos5 (3) → No Change (1 is now permanently in Pos 1)

Second Step (Finding smallest for Pos 2)	
1, [8], [4], 9, 3	Pos2 (8) > Pos3 (4) → Swap → 1, 4, 8, 9, 3
1, [4], 8, [9], 3	Pos2 (4) NOT > Pos4 (9) → No Change
1, [4], 8, 9, [3]	Pos2 (4) > Pos5 (3) → Swap → 1, 3, 8, 9, 4

Third & Fourth Step	
1, 3, [8], 9, [4]	(Step 3): Pos3 (8) > Pos5 (4) → Swap → 1, 3, 4, 9, 8
1, 3, 4, [9], [8]	(Step 4): Pos4 (9) > Pos5 (8) → Swap → 1, 3, 4, 8, 9

Advantages

- Very easy to code.
- Performs fewer swaps overall compared to Bubble Sort.
- Does not need extra memory.

Disadvantages

- Time grows exponentially (nested loops).
- Must finish every single loop (blind to already sorted arrays).
- Not stable (might swap relative positions of equal items).

2.4.2 Linear Search

Also known as sequential search. It works by examining each element in the list one by one until the target is found. It works on both sorted and unsorted data.

اسے سیکوینشل سرچ (Sequential Search) بھی کہا جاتا ہے۔ یہ ایک ایک کر کے فہرست میں ٹارگٹ تلاش کرتا ہے۔ یہ ترتیب شدہ اور غیر ترتیب شدہ دونوں ڈیٹا پر کام کرتا ہے۔

Algorithm

```
Step 1: Start
Step 2: DATA := {8,3,6,1,7,2,4,5}
Step 3: N := 8
Step 4: FindValue := 7
Step 5: ValueFound := 0
Step 6: Repeat for PTR = 1 to N
Step 7: If DATA[PTR] = FindValue:
    a. ValueFound = 1
    b. Goto Step 8
Step 8: If ValueFound = 1:
    Print DATA[PTR]
    Else:
    Print "Not Found"
Step 9: End
```

Pseudocode

```
Start
DATA = {8, 3, 6, 1, 7, 2, 4, 5}
N = 8
FindValue = 7
ValueFound = 0

For PTR = 0 to N-1
    If DATA[PTR] = FindValue then
        ValueFound = 1
        Exit Loop
    End If
End For

If ValueFound = 1 then
    Print DATA[PTR]
Else
    Print "Data Not Found"
End If
End
```

Linear Search Example

Data List: 8, 3, 6, 1, 7, 2, 4, 5

Searching Item: 7

Step	Element Checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not Found
2	3	3 == 7 ?	Not Found
3	6	6 == 7 ?	Not Found
4	1	1 == 7 ?	Not Found
5	7	7 == 7 ?	Found! (Exit Loop)

Advantages

- Does not care if data is scrambled or sorted.
- Extremely simple code (one loop, one if statement).
- Faster for very small lists (no setup time).
- Does not require any extra space.

Disadvantages

- If target is at the end of 1 million items, it does 1 million checks.
- Time grows linearly (double the list = double the time).
- Inefficient if you need to search the same list thousands of times.

2.4.2 Binary Search

★ GOLDEN TOPIC

A highly efficient search algorithm that works by repeatedly dividing the search interval in half. **Crucial limitation: It only works on a SORTED list.**

یہ ایک انتہائی تیز سرچ الگورتھم ہے جو بار بار لسٹ کو دو حصوں میں تقسیم کرتا ہے۔ اہم شرط: یہ صرف ترتیب شدہ (Sorted) فہرست پر کام کرتا ہے۔

Algorithm (Summary)

```
Step 1: Start
Step 2: DATA := {1,2,3,4,5,6,7,8,9} (Sorted)
Step 3: BEG = 1, END = 9
Step 4: MID = INT((BEG+END)/2)
Step 5: Item = 7
Step 6: LOC = 0
Step 7: Repeat while BEG ≤ END AND DATA[MID] ≠ Item
Step 8: If Item < DATA[MID]:
    END = MID - 1
Else:
    BEG = MID + 1
Step 9: MID = INT((BEG+END)/2)
Step 10: [End loop]
Step 11: If DATA[MID] = Item:
    LOC = MID
Else:
    LOC = 0
Step 12: Print Result based on LOC
Step 13: End
```

Pseudocode

```
Start
DATA = {1,2,3,4,5,6,7,8,9}
BEG = 1, END = 9
MID = INT((BEG+END)/2)
Item = 7
LOC = NULL

While BEG ≤ END OR DATA[MID] ≠ Item
    IF Item < DATA[MID] then
        END = MID - 1
    Else
        BEG = MID + 1
    End If
    MID = INT((BEG+END)/2)
End While

If DATA[MID] = Item then
    LOC = MID
Else
    LOC = NULL
End If

If LOC ≥ 0 then
    Print DATA[MID]
Else
    Print "Not Found"
End If
End
```

Binary Search Example

Sorted List: [1, 2, 3, 4, 5, 6, 7, 8]

Searching Item: 3

Step 1: $MID = 8/2 = 4$ (Value: 4). $3 < 4$, go Left.

1	2	3	4	5	6	7	8
			MID				

Step 2: Sub-list [1,2,3]. $MID = 3/2 = 1.5 \rightarrow 2$. (Value: 2). $3 > 2$, go Right.

1	2	3
		MID

Step 3: Sub-list [3]. $MID = 1/2 = 0.5 \rightarrow 1$. (Value: 3). $3 == 3$. FOUND!



Advantages

- Extremely fast for "Big Data". Finds an item in 1M elements in just ~20 steps.
- Massive time savings for static lists searched repeatedly.

Disadvantages

- **Must be sorted first.** Fails on scrambled data.
- Requires random access to elements (works on Arrays, not easily on Linked Lists).
- More complex code, prone to logic errors (off-by-one).

2.5 Evaluation Criteria of Algorithm

2.5.1 Correctness

An algorithm is correct if its output matches expected outcomes, handles edge cases gracefully, and is guaranteed to finish (no infinite loops).

الگورتھم اس وقت درست مانا جاتا ہے جب اس کا آؤٹ پٹ متوقع نتائج کے مطابق ہو اور وہ ختم بھی ہو (لامتناہی لوپ نہ ہو)۔

Trace Table Example (Even/Odd Checker):

Input	Operation (Divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient=1, Remainder=1)	No	Odd
6	6 / 2 (Quotient=3, Remainder=0)	Yes	Even
15	15 / 2 (Quotient=7, Remainder=1)	No	Odd

2.5.2 Efficiency

We measure algorithm efficiency through time (speed) and space (memory). How fast does it run as input size grows? Does it scale well?

ہم الگورتھم کی کارکردگی کو وقت (Speed) اور میموری (Space) کے ذریعے مانتے ہیں۔ ڈیٹا بڑھنے کے ساتھ اس کی رفتار کیسی رہتی ہے؟

2.5.3 Clarity (Readability)

Algorithm must be easy to read, understandable and easy to modify. It should use clear steps, meaningful variables, and good indentation.

الگورتھم کو پڑھنے، سمجھنے اور تبدیل کرنے میں آسان ہونا چاہیے۔ واضح مراحل اور اچھے متغیرات (Variables) کا استعمال ضروری ہے۔

2.6 How to Decide Which Algorithm to Use

Before choosing an algorithm, ask: What is the input size? Is the data sorted? Do we need speed or memory efficiency?

Simple Decision Guide

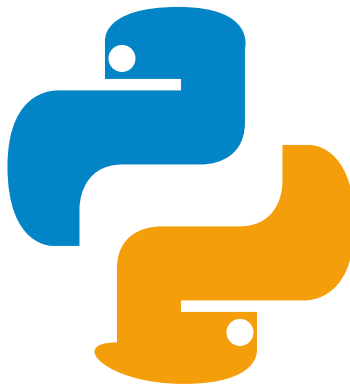
- If searching on **unsorted** data → use **Linear Search**.
- If searching on **sorted** data → use **Binary Search**.
- If data needs **ordering** → use **Sorting algorithm**.

Summary

- **Computational Thinking:** Systematic problem solving (Decomposition, Pattern Recognition, Abstraction, Algorithms).
- **Algorithm:** Step-by-step logical procedure (conceptual).
- **Pseudocode:** Human-readable structured code.
- **Decomposition:** Breaking big problems into smaller parts.
- **Abstraction:** Hiding noise, focusing on essentials.
- **Bubble/Selection Sort:** Simple, nested-loop sorting algorithms (slow for large data).
- **Linear Search:** Checks one by one (works on unsorted data).
- **Binary Search:** Splits in half (extremely fast, but requires sorted data).
- **Evaluation Criteria:** Correctness, Efficiency (Time/Space), and Clarity.

COMPUTER SCIENCE XI

CHAPTER 3: PROGRAMMING FUNDAMENTALS



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

یہ نوٹس خاص طور پر اساتذہ کے لیے تیار کیے گئے ہیں۔ حوالہ کے لیے نصابی کتاب کی پی ڈی ایف فائل "CS XI (3 & 4).pdf" کا استعمال کریں، جس کی من و عن پیروی کی گئی ہے۔ جن موضوعات پر ★ کا نشان ہے وہ امتحانات کے لیے انتہائی اہم ہیں۔

These notes are prepared specifically for teachers. For reference, please consult the textbook PDF file named exactly "**CS XI (3 & 4).pdf**", which has been followed verbatim. Topics marked with a ★ are highly critical for examinations.

3.1 Programming

Computers are dumb machines. They cannot think or do anything without instructions. A set of instructions is called a **program**, and the language used to give these instructions is a **programming language**.

کمپیوٹر ایک بے وقوف مشین ہے۔ یہ ہدایات کے بغیر کچھ نہیں کر سکتا۔ ہدایات کے مجموعے کو پروگرام کہتے ہیں، اور جس زبان میں یہ ہدایات دی جاتی ہیں اسے پروگرامنگ لینگویج کہتے ہیں۔

3.1.1 Programming Language Types

Programming languages are classified into three main types:

Type	Description	Examples
Low Level مشین کے قریب	Closer to the native language of the computer (binary 0s & 1s). Very difficult for humans to understand.	Machine Language (Binary)
Mid Level درمیانی سطح	Bridges the gap. Has hardware access capabilities but is easier to read than low-level code. Uses mnemonics.	Assembly Language
High Level انسان کے قریب	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



Common High-Level Languages

C++

Powerful, closer to hardware. Used for system software, operating systems, and high-performance computer games.

Java

"Write once, run anywhere". Used for large enterprise applications, Android mobile apps, and server programming.

Python

Simple, readable syntax. Widely used for Data Science, AI, Web Development, and by beginners.

3.2 Career Growth with Python

پاتھوں سیکھنے کے بعد مستقبل میں کئی شاندار مواقع (Career Opportunities) موجود ہیں کیونکہ یہ سیکھنے میں آسان اور بہت طاقتور ہے۔

- **Software Development:** Build desktop and web applications.
- **Data Science and Data Analysis:** Analyze data, create reports, and make predictions.
- **Artificial Intelligence and Machine Learning:** Develop intelligent systems, chatbots, and recommendation algorithms.
- **Web Development:** Use frameworks like Django and Flask to build websites.
- **Automation and Scripting:** Automate repetitive office or lab tasks.
- **Cybersecurity and Networking:** Write security tools and analyze network data.

💡 Why Python?

Remind students that Python allows them to focus on **problem-solving and logic** rather than memorizing complex syntax rules.

3.3 Integrated Development Environment (IDE)

An IDE is a software tool that helps programmers write, run, and manage code in one place. It is a complete workspace for programming.

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جو پروگرام لکھنے، چلانے اور سنبھالنے میں مدد دیتا ہے۔ یہ پروگرامنگ کے لیے ایک مکمل ورک اسپیس ہے۔

Key Features of an IDE:

- **Code Editor:** Where you write and edit code.
- **Syntax Highlighting:** Colors the code to make it easier to read.
- **Auto completion:** Suggests code while you type.
- **Run/Execute Tool:** Executes the program.
- **Debugger:** Helps find and fix errors.

3.3.1 Visual Studio Code (VS Code)

VS Code is a free, lightweight, and powerful code editor widely used for Python. It is not a full IDE by default, but it can be turned into one by installing extensions.



VS Code Interface Elements

Element	Description
1. Top Menu Bar	Control over files, editing, navigation, and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains core tools: Explorer, Search, Source Control, Run & Debug, and Extensions.
3. Explorer	Shows project files and folders. Helps create, open, and manage your Python files.
4. Search Bar	Finds and replaces text across your currently open file or entire project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes, and collaborate.
6. Run & Debug	Allows you to execute code, find errors, pause at specific lines, and inspect variables.
7. Extensions	Like an App Store for VS Code. Add features like the Python extension, Prettier, or Python Debugger.



Teacher's Demo Setup

Ensure you have the **Python extension by Microsoft** installed in VS Code before class to demonstrate syntax highlighting and the Run button.

3.4 Fundamentals of Python Programming

★ GOLDEN TOPIC

The basic structure of a Python program is simple and clear. It processes data using statements, and finally shows output. Programs run top-to-bottom sequentially.

پائتھون پروگرام کی ساخت بہت سادہ ہے۔ اس میں کوڈ اوپر سے نیچے (Top to Bottom) چلتا ہے۔

1. Comments (#)

Comments are notes written in the code for humans. They start with the # symbol. The Python interpreter completely ignores them.

کمنٹس (Comments) کوڈ کو سمجھانے کے لیے لکھے جاتے ہیں اور پائتھون انہیں نظر انداز کر دیتا ہے۔

2. Indentation (Spaces)

Unlike C++ or Java which use curly brackets {} to define blocks of code, Python uses **indentation** (spaces at the beginning of a line). Indentation is mandatory! The standard is 4 spaces (or 1 tab).

```
# This is a comment. It explains the code below.  
print("Hello, World!")  
  
if 5 > 2:  
    # The space before 'print' is the Indentation  
    print("Five is greater than two!")
```

⚠ Common Error

If a student gets an IndentationError, tell them to check their spaces. Mixing tabs and spaces can break a Python program.

3.4.1 Variables

★ GOLDEN TOPIC

A variable is a named space in computer memory used to store data. Think of it as a container. In Python, you do not need to declare the type; Python figures it out automatically.

ویری ایبل (Variable) میموری میں ایک جگہ کا نام ہے جہاں ڈیٹا محفوظ کیا جاتا ہے۔ یہ ایک ڈبے (Container) کی طرح ہے۔

Rules for Naming Variables:

- Can contain letters, numbers, and underscores (_).
- **MUST start** with a letter or underscore, NOT a number. (e.g., 1name is invalid).
- Are **case-sensitive** (age and Age are completely different).
- Cannot use Python **keywords** (like if, for, while).

Valid Names ✓

```
name = "Ali"  
age = 16  
my_height = 5.7  
score1 = 100
```

Invalid Names ✗

```
1name = "Ali"    # starts with  
number  
my height = 5.7  # contains space  
if = 100         # keyword used
```

Data Types in Python

Data types specify the kind of data a variable holds. Python has several built-in data types:

Name	Type	Description & Example
Integers	int	Whole numbers without decimals. age = 16
Floating Point	float	Numbers with a decimal point. height = 5.7
Strings	str	Text data, enclosed in quotes. name = "Sammy" or year = "2026"
Boolean	bool	Logical value indicating True or False (Capital T/F). is_student = True
Lists	list	Ordered, changeable collection [brackets]. marks = [85, 90, 75]
Tuples	tup	Ordered, unchangeable sequence (parentheses). coords = (10, 20)
Dictionaries	dict	Unordered Key:Value pairs {curly braces}. student = {"name": "Ahmed", "age": 16}
Sets	set	Unordered collection of unique objects. vowels = {'a', 'e', 'i'}

3.4.2 Input and Output Handling

Output: The print() function

Displays information on the screen.

```
# 1. Print only a string
print("Hello Class!")

# 2. Print a string and a variable (comma separates them)
name = "Ali"
print("Welcome,", name)

# 3. f-strings (Fastest & most readable way to format)
age = 16
print(f"My name is {name} and I am {age} years old.")
```

Input: The input() function

Pauses the program and waits for the user to type something. **Whatever the user types is always treated as a string (text)** by default.

یوزر جو بھی ٹائپ کرتا ہے، پائتھون اسے ہمیشہ String (ٹیکسٹ) سمجھتا ہے۔ اگر نمبر چاہیے تو اسے تبدیل (Convert) کرنا پڑتا ہے۔

```
# Taking text input
user_name = input("Enter your name: ")

# Type Casting: Converting string input to an integer (int)
age = int(input("Enter your age: "))

# Type Casting: Converting string input to a decimal (float)
height = float(input("Enter your height: "))
```

3.4.3 Operators and Operands

★ GOLDEN TOPIC

An **Operand** is the value or variable. An **Operator** is the symbol that performs the action.

10 (Operand) + (Operator) 20 (Operand)

1. Arithmetic Operators (Math)

Operator	Purpose	Example	Result
+	Addition	5 + 3	8
-	Subtraction	10 - 4	6
*	Multiplication	4 * 2	8
/	Division (always float)	9 / 2	4.5
//	Floor division (drops decimal)	9 // 2	4
%	Modulus (remainder)	9 % 2	1
**	Exponentiation (power)	2 ** 3	8

2. Assignment Operators

Used to assign values to variables, often updating them.

Operator	Example	Equivalent To
=	x = 5	Assign 5 to x
+=	x += 2	x = x + 2
-=	x -= 1	x = x - 1
*=	x *= 3	x = x * 3

Other Important Operators

3. Relational (Comparison) Operators

Compare two values and return a Boolean (True or False).

Op	Description	Example (If a=10, b=5)	Result
==	Equal to	a == b	False
!=	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
>=	Greater/Equal	a >= 10	True

4. Logical Operators

Combine multiple conditions.

Operator	Description
and	True ONLY if both conditions are true.
or	True if at least one condition is true.
not	Reverses the result (True becomes False).

5. Membership Operators

Check if a value exists inside a sequence (like a string or list).

```
name = "Python"
print("P" in name)      # Output: True
print("z" not in name)  # Output: True
```

6. Bitwise Operators

Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and perform operations bit by bit. They act like switches turning bits ON (1) or OFF (0). Python uses the `bin()` function to see the binary form of a number.

بٹوائز آپریٹرز نمبرز کو بائنری (0 اور 1) میں تبدیل کر کے ان پر کام کرتے ہیں۔

Bitwise AND (&)

Returns 1 ONLY if BOTH corresponding bits are 1. Otherwise, 0.

Bit A	Bit B	A & B
0	0	0
0	1	0
1	0	0
1	1	1

Bitwise OR (|)

Returns 1 if AT LEAST ONE of the bits is 1.

Bit A	Bit B	A B
0	0	0
0	1	1
1	0	1
1	1	1

Example: (a = 10, b = 6)

```
# a = 10 (Binary: 1010)
# b = 6 (Binary: 0110)

print(10 & 6) # AND: 0010 (Decimal 2)
print(10 | 6) # OR : 1110 (Decimal 14)
```

Bitwise Operators (Continued)

Bitwise XOR (^)

Returns 1 if bits are DIFFERENT. Returns 0 if they are the SAME.

Bit A	Bit B	A ^ B
0	0	0
0	1	1
1	0	1
1	1	0

Bitwise NOT (~)

Inverts ALL bits. 0 becomes 1, 1 becomes 0. Also adds negative sign minus 1.

Bit A	~A
0	1
1	0

~10 gives -11

Bitwise Shifts

Left Shift (<<): Shifts bits left. Multiplies by 2 for each shift.

Right Shift (>>): Shifts bits right. Divides by 2 (floor) for each shift.

```
a = 5          # Binary: 0000 0101

print(a << 1) # Left Shift 1: 0000 1010 (Decimal 10) - Adds 0 on right
print(a << 2) # Left Shift 2: 0001 0100 (Decimal 20) - Adds 00 on right

b = 20         # Binary: 0001 0100
print(b >> 1) # Right Shift 1: 0000 1010 (Decimal 10) - Removes rightmost
```

3.5 Control Structures

★ GOLDEN TOPIC

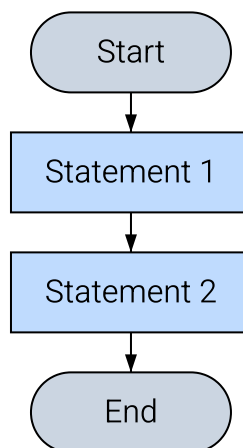
Control structures decide the order in which statements are executed in a program. Almost every program uses three basic structures: **Sequence**, **Selection**, and **Repetition**.

کنٹرول اسٹرکچر یہ طے کرتے ہیں کہ پروگرام کی لائنیں کس ترتیب سے چلیں گی۔

3.5.1 Sequence

Sequence means that statements are executed one after another, strictly from top to bottom, in the exact order they are written. This is the default behavior of Python.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b                          # Step 3: Process
print("Sum =", total)                  # Step 4: Output
```



3.5.2 Selection Statements

Selection alters the flow based on a condition (True/False). It helps programs respond intelligently.

1. The if Statement

Executes a block of code ONLY if the condition is True. If False, the block is completely skipped.

یہ کوڈ صرف اسی وقت چلے گا جب شرط (Condition) سچ ہو۔ ورنہ یہ حصہ نظر انداز ہو جائے گا۔

```
age = int(input("Enter your age: "))

if age ≥ 18:
    print("You are an Adult.") # Runs only if ≥ 18

print("Program continues ... ") # Runs every time
```

2. The if-else Statement

Provides TWO paths. IF true, do this. ELSE, do that. One of the blocks will *always* execute.

اس میں دو راستے ہوتے ہیں۔ اگر شرط سچ ہے تو پہلا حصہ چلے گا، ورنہ (else) دوسرا حصہ لازمی چلے گا۔

```
age = int(input("Enter your age: "))

if age ≥ 18:
    print("Apply for CNIC.")
else:
    print("Apply for CRC (B-Form).")
```

Formatting Rule

Don't forget the colon (:) after if and else, and ensure the code inside the blocks is indented!

Selection Statements (Continued)

3. The if-elif-else Statement

Handles multiple conditions. It checks from top to bottom. As soon as it finds a True condition, it executes that block and ignores the rest.

```
marks = int(input("Enter marks: "))

if marks ≥ 80:
    print("A-1 Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 60:
    print("B Grade")
else:
    print("Fail or below B")
```

4. Nested if Statement

An if inside another if. Used when a decision requires passing multiple levels of checks.

```
age = 20
salary = 35000

if age ≥ 18:
    # Passed first check
    if salary ≥ 30000:
        # Passed second check
        print("Loan APPROVED!")
    else:
        print("Salary too low for loan.")
else:
    print("Too young for loan.")
```

3.5.3 Repetition Statements (Loops)

★ GOLDEN TOPIC

A repetition statement (loop) allows a block of code to be executed repeatedly. This drastically reduces the need to write the same code over and over.

لوپس (Loops) کسی کام کو بار بار دہرانے کے لیے استعمال ہوتے ہیں۔

1. The for Loop & range() function

Used when the **exact number of iterations is known**. The range() function generates the sequence of numbers for the loop.

Syntax	How it works	Code Example & Output
range(stop)	From 0 up to (but NOT including) the stop value.	for i in range(3): print(i) Output: 0 1 2
range(start, stop)	From start up to (NOT including) stop.	for i in range(2, 5): print(i) Output: 2 3 4
range(start, stop, step)	From start to stop, jumping by step amount.	for i in range(1, 6, 2): print(i) Output: 1 3 5

Example: Even & Odd Loop

```
for i in range(0, 6, 2):  
    print("Even:", i)
```

Output:

Even: 0

Even: 2

Even: 4

Repetition Statements (Continued)

2. The while Loop

Repeats a block of code **as long as a condition remains True**. Used when the exact number of iterations is NOT known in advance.

یہ لوپ اس وقت تک چلتا رہے گا جب تک اس کی شرط سچ (True) رہے گی۔

```
count = 1
while count ≤ 3:
    print(count)
    count += 1    # VERY IMPORTANT: Update condition!

# Output: 1 2 3
```

Loop Control Statements

These statements alter the normal flow of a loop.

break

Exits/Stops the loop entirely immediately.

```
for i in range(5):
    if i == 3:
        break
    print(i)
# Stops printing at 2
```

continue

Skips current iteration and jumps to the next.

```
for i in range(5):
    if i == 3:
        continue
    print(i)
# Skips 3 (Prints 0 1 2 4)
```

Nested Loops & Comparison

Nested Loop

A loop inside another loop. The inner loop runs *completely* from start to finish for *every single iteration* of the outer loop.

ایک لوپ کے اندر دوسرا لوپ۔ باہر والا لوپ ایک بار چلتا ہے، تو اندر والا لوپ اپنا پورا چکر مکمل کرتا ہے۔

```
for outer in range(1, 3):          # Runs 2 times (1, 2)
    for inner in range(1, 4):      # Runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

Output:

```
Outer: 1, Inner: 1
Outer: 1, Inner: 2
Outer: 1, Inner: 3
Outer: 2, Inner: 1
Outer: 2, Inner: 2
Outer: 2, Inner: 3
```

for Loop vs while Loop

for Loop	while Loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the counter.	Requires manual update of counter (risk of infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic scenarios (like "keep playing until user says Stop").

Practical Activity: Shopping Bill Calculator

This program beautifully combines Sequence, Selection, and Repetition.

```
print("===== SHOPPING BILL =====")
total = 0.0
count = 0
shopping = True

# REPETITION: Keep adding items
while shopping:
    item = input("Enter Item Name (or 'done' to stop): ")

    # SELECTION: Check if user wants to stop
    if item.lower() == "done":
        shopping = False
    else:
        price = float(input(f"Enter Price of {item}: "))

        # SELECTION: Validate price
        if price <= 0:
            print("Invalid price!")
        else:
            total += price
            count += 1

# Summary & Discount
if total >= 5000:
    discount = total * 0.20 # 20% off
else:
    discount = 0

final = total - discount
print(f"Total Items: {count}")
print(f"Total Bill: {total}")
print(f"Final to Pay: {final}")
```

3.6 Libraries in Python

Libraries are pre-written code that offer helpful functions. Instead of writing complex code from scratch, programmers use libraries to solve common problems efficiently.

لائبریری پہلے سے لکھا ہوا کوڈ ہوتا ہے جسے ہم اپنے پروگرام میں استعمال کر سکتے ہیں تاکہ ہمیں سب کچھ خود سے نہ لکھنا پڑے۔

How to Import Libraries

There are mainly 5 ways to import a library:

Syntax	Code Example
1. Import entire module Must use <code>module.func()</code>	<pre>import math res = math.sqrt(25)</pre>
2. Import with alias (nickname)	<pre>import math as m res = m.sqrt(16)</pre>
3. Import specific function No need to prefix module name.	<pre>from math import sqrt res = sqrt(36)</pre>
4. Import multiple functions	<pre>from math import sqrt, pi</pre>
5. Import everything (*)	<pre>from math import * res = sqrt(64)</pre>

Best Practice

Advise students that importing specific functions (method 3) is better for memory and clarity than importing everything using the `*` wildcard.

3.6.1 Built-in Libraries

These libraries come pre-installed with Python. You just need to import them to use them.

1. math Library

Provides mathematical operations and constants.

```
import math

print(math.pi)           # 3.14159...
print(math.ceil(4.3))    # 5 (Round UP)
print(math.floor(4.9))   # 4 (Round DOWN)
print(math.pow(2,4))      # 16.0 (2 to
                           # power 4)
print(math.sqrt(81))     # 9.0
```

2. random Library

Generates random numbers or selections.

```
import random

# Random int between 1 and 10
num = random.randint(1, 10)
print(num)

# Random float between 0.0 to 1.0
print(random.random())
```

3. datetime Library

Provides classes for working with dates and times.

```
import datetime

# Get exact current date and time
now = datetime.datetime.now()
print(now) # e.g., 2026-09-27 11:59:40

# Format it beautifully (Day-Month-Year)
formatted = now.strftime("%d-%m-%Y")
print(formatted) # e.g., 27-09-2026
```

3.6.2 Third-Party Libraries

Third-party libraries are NOT included with Python by default. They are built by the community and must be installed via terminal using: `pip install library_name`.

یہ پائتھون میں پہلے سے موجود نہیں ہوتیں۔ انہیں pip کمانڈ کے ذریعے انسٹال کرنا پڑتا ہے۔

Library Name	Primary Use
NumPy	Heavy numerical computing and large arrays.
Pandas	Data analysis, handling table-like data (Excel/CSV).
Matplotlib	Data visualization, plotting graphs and charts.
TensorFlow / PyTorch	Artificial Intelligence and Machine Learning.

3.7 Debugging Code in Python

Debugging is the process of identifying, tracing, and resolving issues (bugs) within a program so it works correctly.

پروگرام میں موجود غلطیوں (Bugs) کو تلاش کرنا اور انہیں ٹھیک کرنا ڈیباگنگ کہلاتا ہے۔

Types of Errors (Bugs)

1. Syntax Errors

Occur when the **rules (grammar) of Python are violated**. The program will NOT run at all until it is fixed.

ٹائپنگ کی غلطی یا پائتھون کے گرامر کی خلاف ورزی (جیسے بریکٹ بند نہ کرنا)۔

```
print("Hello World" # Missing closing parenthesis!
```

2. Runtime Errors (Exceptions)

The code is written correctly, so it starts running. But an impossible operation crashes it **during execution**.

پروگرام چلنا شروع ہو جاتا ہے، لیکن چلتے ہوئے کسی ناممکن کام (جیسے صفر سے تقسیم) کی وجہ سے کریش ہو جاتا ہے۔

```
x = 10 / 0 # ZeroDivisionError: division by zero!
```

3. Logical Errors

The hardest to find. The program runs perfectly without crashing, but it produces the **wrong output** because the programmer's logic is flawed.

کوئی ایرر میسج نہیں آتا اور پروگرام پورا چلتا ہے، لیکن جواب غلط آتا ہے کیونکہ آپ کی منطق (Logic) غلط تھی۔

```
area = length + width # Logical error: Should be multiplied (*), not added.
```

Reading Error Messages (Tracebacks)

When Python crashes, it prints a **Traceback**. It is a roadmap to fix the bug. It tells you:

1. What type of error occurred.
2. Which file the error is in.
3. Exactly **which line number** caused the crash.

```
Traceback (most recent call last):
  File "e:\Python Code\debug_program.py", line 1, in <module>
    x = int("abc")
ValueError: invalid literal for int() with base 10: 'abc'
```

Diagnosis: The traceback clearly states that on Line 1, we tried to convert letters ("abc") into an integer, which generated a ValueError.

Summary

- **Python** is a High-Level, readable language great for AI, Data, and Web.
- **VS Code** is a popular editor made into an IDE via Extensions.
- **Variables** do not need explicit types in Python; types change dynamically.
- **Data Types:** int, float, str, bool, list, dict.
- **Input/Output:** print() outputs data; input() gets data *as a string*.
- **Control Structures:** Sequence (top-to-bottom), Selection (if/else), Repetition (for/while).
- **Libraries:** Extend Python's power. Built-in (math, random) or Third-party (NumPy).
- **Errors:** Syntax (Grammar), Runtime (Crash during run), Logical (Wrong output).

COMPUTER SCIENCE XI

CHAPTER 4: DATA AND ANALYSIS



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

یہ نوٹس خاص طور پر اساتذہ کے لیے تیار کیے گئے ہیں۔ حوالہ کے لیے نصابی کتاب کی پی ڈی ایف فائل "CS XI (3 & 4).pdf" کا استعمال کریں، جس کی من و عن پیروی کی گئی ہے۔ جن موضوعات پر ★ کا نشان ہے وہ امتحانات کے لیے انتہائی اہم ہیں۔

These notes are prepared specifically for teachers. For reference, please consult the textbook PDF file named exactly "**CS XI (3 & 4).pdf**", which has been followed verbatim. Topics marked with a ★ are highly critical for examinations.

4.1 Concepts of Database and Elements

4.1.1 Data, Information and Database

Data (ڈیٹا)

Data are unprocessed facts, figures, or observations collected from different sources. It can be quantitative (numerical) or qualitative (descriptive).

ڈیٹا سے مراد خام (Unprocessed) حقائق اور اعداد و شمار ہیں۔ یہ بذات خود کوئی مکمل مطلب نہیں دیتے۔

- Example: 80, 75, 90, 70, 85
- Example: The color of cars in a parking lot.

Information (معلومات)

When we organize, process, and interpret data, it becomes information which provides meaning and supports decision-making.

جب ڈیٹا کو پروسیس اور منظم کیا جاتا ہے تاکہ اس کا کوئی مطلب نکلے، تو وہ معلومات (Information) کہلاتا ہے۔

- Example: The average score of the class is 80%.



Aspect	Data	Information
Definition	Raw facts and figures	Processed and Meaningful data
Example	85, 90, 75	Average Marks = 83.3%
Usefulness	Less useful on its own	Useful for decision making

Database and DBMS

Database (ڈیٹا بیس)

A database is an organized collection of related data that is stored in a structured manner so that it can be easily accessed, managed, and updated. In traditional systems, data was stored in files or books, making it difficult to search. A database solves these problems.

ڈیٹا بیس متعلقہ ڈیٹا کا ایک منظم مجموعہ ہے جسے اس طرح محفوظ کیا جاتا ہے کہ اسے آسانی سے تلاش، منظم اور اپ ڈیٹ کیا جاسکے۔

4.1.2 Database Management System (DBMS)

A DBMS is software that allows users to create, manage, and interact with databases. It acts as a bridge between the user and the database. Instead of directly handling data files, users interact with DBMS.

یہ ایک سافٹ ویئر ہے جو صارفین کو ڈیٹا بیس بنانے اور سنبھالنے میں مدد دیتا ہے۔ (جیسے، MS Access، MySQL، Oracle)۔

Advantages of DBMS:

1. Less Redundancy (کم تکرار)

The same data is not stored multiple times unnecessarily. E.g., Department name is stored once, not in every student record.

2. Improved Consistency (مستقل مزاجی)

Ensures all users see the same correct data. Updating a phone number once updates it everywhere.

3. Improved Security (بہتر سیکیورٹی)

Allows only authorized users to access sensitive info. (Admin sees marks, staff sees only names).

4. Data Integrity (ڈیٹا کی درستگی)

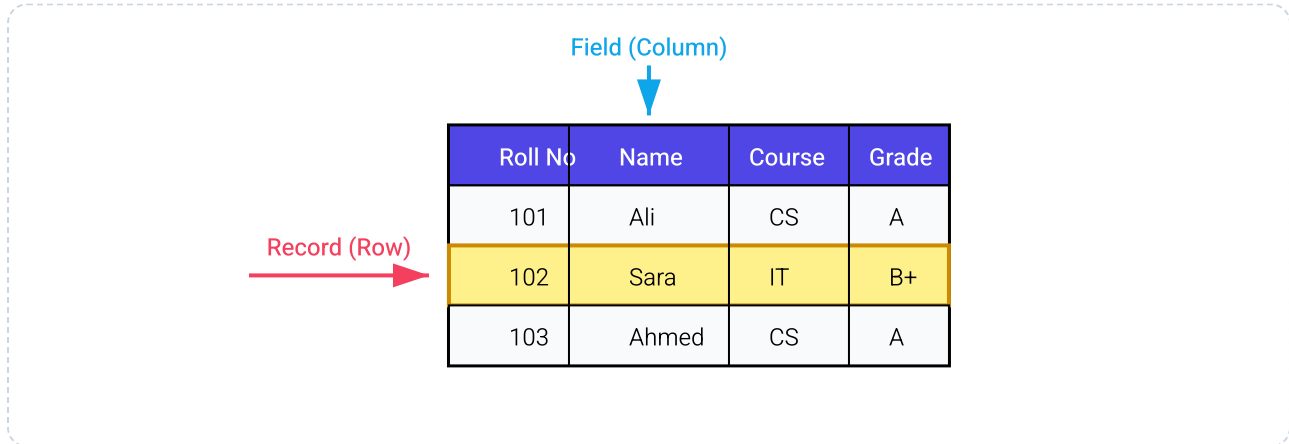
Ensures stored data is accurate and reliable. (E.g., preventing duplicate Roll Numbers).

4.2 Database Components

A database is made up of several important components that help organize data effectively. The most basic structure is a table.

ایک ڈیٹابیس کئی اہم حصوں سے مل کر بنتا ہے۔ اس کا سب سے بنیادی حصہ ٹیبل (Table) ہے۔

Component	Description
Table (ٹیبل)	Stores data in rows and columns, similar to a spreadsheet. Each table stores information about a specific entity (e.g., Students, Teachers).
Record / Row (ریکارڈ / قطار)	A complete set of related data about ONE specific instance. (e.g., One entire row containing Ali's Name, Roll No, and Marks).
Field / Column (فیلڈ / کالم)	A single piece of information or attribute. (e.g., The entire column that holds ONLY "Names" or ONLY "Marks").



4.3.1 Keys and Integrity Constraints

★ GOLDEN TOPIC

A key field is an attribute (or set of attributes) that helps uniquely identify records and link tables together. Understanding keys is essential for designing efficient databases.

کی فیلڈز (Key Fields) وہ کالم ہوتے ہیں جو ریکارڈز کو منفرد شناخت (Unique ID) دیتے ہیں اور مختلف ٹیبلز کو آپس میں جوڑتے ہیں۔

1. Candidate Key (امیدوار کیز)

Any field (or set of fields) that *has the potential* to uniquely identify a record. A table may have multiple candidate keys, but only ONE is selected as the primary key.

- Must contain unique values.
- Cannot contain NULL (empty) values.
- *Example:* In an Employee table: EmpID, EmpLicense#, and EmpPassport are all Candidate Keys.

2. Primary Key (پرائمری کی)

The **one specific field** chosen from the Candidate Keys to uniquely identify every record in the table. It ensures no two records are exactly alike.

- Must contain unique values.
- Cannot contain NULL values.
- Each table has ONLY ONE Primary Key.
- *Example:* We select EmpID as the Primary Key.

💡 Analogy

In a classroom, students have a Roll Number, a B-Form Number, and a Phone Number. All three are **Candidate Keys**. The school chooses Roll Number as the main identifier, making it the **Primary Key**.

Keys (Continued) & RDBMS

3. Alternate Key (متبادل کیز)

The Candidate Keys that were **NOT selected** as the Primary Key. They still hold unique values but are left as alternates.

- Derived from Candidate Keys.
- *Example:* Since EmpID became the Primary Key, EmpLicense# and EmpPassport become Alternate Keys.

4. Foreign Key (فارن کی)

A field in one table that **refers to the Primary Key of another table**. It is used to create relationships (links) between tables.

- CAN contain duplicate values (e.g., many employees in one department).
- CAN contain NULL values (in some cases).
- *Example:* Dept ID in the Employee table links to the Primary Key Dept ID in the Department table.

4.3 Relational Database Management System (RDBMS)

An RDBMS stores data in multiple tables and uses keys (like Primary and Foreign keys) to maintain relationships between those tables. It is powerful, structured, and used by banks and universities.

آرڈی بی ایم ایس (RDBMS) ڈیٹا کو مختلف ٹیبلز میں محفوظ کرتا ہے اور کیز (Keys) کی مدد سے ان کا آپس میں تعلق (Relationship) قائم کرتا ہے۔

- Reduces data redundancy (duplication).
- Supports multi-user access securely.
- Easy to query using SQL (Structured Query Language).

4.4 Entity Relationship Model (ER-Model)

★ GOLDEN TOPIC

The ER Model is a conceptual tool used to design databases before they are created. It helps designers understand what data needs to be stored and how different pieces are related. It acts like an architect's blueprint for the database.

ای آر ماڈل (ER Model) ڈیٹا بیس بنانے سے پہلے کا ایک خاکہ (Blueprint) ہے۔ یہ بتاتا ہے کہ کون سا ڈیٹا محفوظ کرنا ہے اور ان کا آپس میں کیا تعلق ہے۔

Components of an ER Model

Component	Description & Symbol
1. Entities (اشیاء)	Any real-world object or concept about which data is stored. (e.g., Student, Teacher, Course). Symbol: Rectangle 
2. Attributes (خصوصیات)	The properties or characteristics of an entity. (e.g., StudentID, Name, Class). Symbol: Oval  (Primary Key attributes have an underline)
3. Relationships (تعلق)	How entities are connected to each other. (e.g., A Student "Enrolls in" a Course). Symbol: Diamond 

Types of Relationships in ER Model

Understanding relationships is essential for designing efficient and accurate databases. There are three main types:

1. One-to-One (1:1)

One record is related to ONLY ONE record in another entity. This is rare and often used for security.

Example: One Student has One ID Card.

STUDENT (1) — HAS — (1) ID CARD

2. One-to-Many (1:M)

One record in Entity A relates to MULTIPLE records in Entity B. This is the most common relationship.

Example: One Teacher teaches Many Students.

TEACHER (1) — TEACHES — (N) STUDENT

3. Many-to-Many (M:N)

Multiple records in Entity A relate to multiple records in Entity B.

Example: Many Students enroll in Many Classes.

STUDENT (M) — ENROLLS — (N) CLASS

⚠ The M:N Database Rule

Relational databases **cannot directly support** Many-to-Many relationships. It causes data duplication. To fix this, we must break it into two 1:M relationships using a **Junction Table** (e.g., an "Enrollment" table).

ریشنل ڈیٹابیس میں Many-to-Many کا تعلق براہ راست نہیں بن سکتا۔ اس کے لیے ہمیں درمیان میں ایک نیوٹیل (Junction Table) بنانا پڑتا ہے۔

4.5 Referential Integrity

★ GOLDEN TOPIC

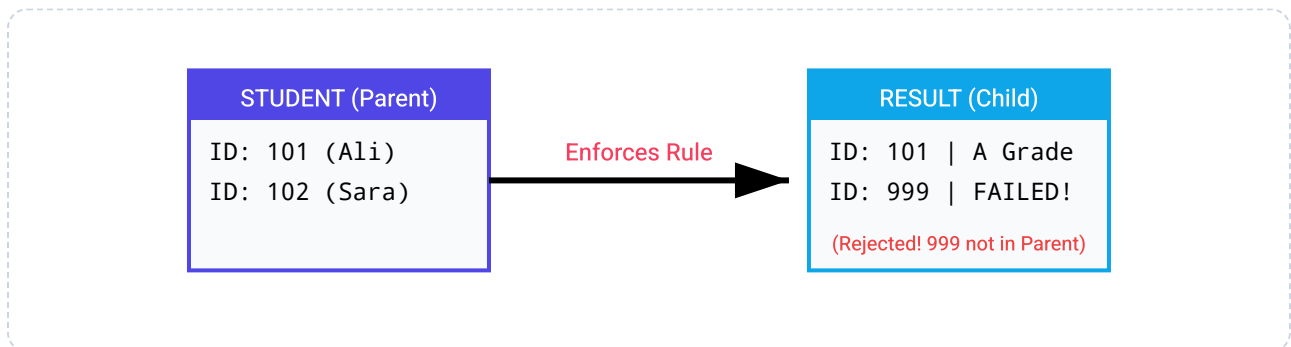
Referential integrity is a fundamental concept that ensures the accuracy and consistency of data across related tables. It ensures that a Foreign Key in one table **always refers to an existing and valid Primary Key** in another table.

ریفرنیشل انٹیگریٹی (Referential Integrity) اس بات کو یقینی بناتی ہے کہ اگر دو ٹیبلز آپس میں جڑے ہوئے ہیں، تو ان کا ڈیٹا درست اور مستقل رہے۔ یعنی فارن کی (Foreign Key) میں ہمیشہ وہی ویلیو ہونی چاہیے جو پرائمری کی (Primary Key) میں موجود ہو۔

Why is it Important?

Imagine a **Student Table** and a **Result Table**. The Result table uses StudentID as a Foreign Key. If Referential Integrity is turned ON:

- You **cannot** enter a result for StudentID "999" if "999" does not exist in the main Student table. (Prevents orphan records).
- You **cannot** delete a student from the Student table if they still have marks in the Result table, unless you handle the related records first.



Referential Integrity Operations

To maintain consistency automatically without manual effort, DBMS provides two powerful operations.

1. Cascade Update

If a Primary Key value is changed in the parent table, that exact change is **automatically updated in all related child tables**.

Example: If Ali's StudentID changes from 101 to 201 in the Student table, his ID will automatically change to 201 in the Enrollments table. No data is lost or disconnected.

2. Cascade Delete

If a record is deleted from the parent table, **all related records in child tables are also deleted automatically**.

Example: If student Sara (ID 102) leaves the school and is deleted from the Student table, all her records in the Enrollments and Results tables are instantly wiped out. This prevents "ghost" data from clogging the system.

Benefits of Referential Integrity	
Maintains complete data consistency.	Saves time and manual effort for admins.
Reduces errors and prevents orphan records.	Ensures reliable and accurate reporting.

4.6 Relational Schema Development

A relational schema is a formal representation of the structure of a database. It acts as the **design or blueprint** of the database before it is actually implemented in software like MS Access.

ریشنل سکیم (Relational Schema) ڈیٹابیس کا ایک خاکہ یا نقشہ ہے جو بتاتا ہے کہ کون سے ٹیبلز بنیں گے اور ان کا آپس میں کیا تعلق ہوگا۔

What does a Schema define?

- How data is organized into Tables.
- What Fields (columns) each table contains.
- Which fields are Primary Keys and Foreign Keys.
- How different tables are related to each other.

Why develop a Schema?

Developing a relational schema is an important step because:

1. It reduces redundancy (duplication of data).
2. It improves data integrity by planning keys in advance.
3. It makes the database easier to manage and scale in the future.

Teacher's Tip

Remind students that skipping the schema design phase is like trying to build a house without an architectural drawing. It leads to messy, broken structures.

4.7 Case Study: Library Management System

This case study demonstrates the step-by-step development of an ER schema, from requirement analysis to the final ER diagram for a college library.

Step 1: Requirement Analysis

What does the system need to do?

- A book has an ID, title, author, and publisher.
- A member (student/teacher) has an ID, name, and contact info.
- A member can borrow multiple books; a book can be borrowed by different members over time.
- The system must record Issue Date and Return Date.

Step 2: Identification of Entities

From the requirements, we identify two main real-world objects (Entities):

1. **Book**
2. **Member**

Step 3: Identification of Attributes

Book Entity	Member Entity
• BookID (Primary Key) • Title • Author • Publisher	• MemberID (Primary Key) • Name • ContactInfo

Library Case Study (Continued)

Step 4: Identification of Relationships

We need to understand how the entities interact. The primary interaction is:
"A Member *borrow*s a Book"

Step 5: Determine Cardinality

Cardinality defines the nature of the relationship (1:1, 1:M, M:N).

- A single member can borrow **many** books over time.
- A single book can be borrowed by **many** different members over time.

Conclusion: The relationship is Many-to-Many (M:N)



Library Case Study (Continued)

Step 6: Resolving Many-to-Many Relationship

As discussed earlier, relational databases cannot directly support M:N relationships. It leads to data redundancy. We must resolve it by introducing a **Junction Entity (Bridge Table)**.

ریشنل ڈیٹابیس میں Many-to-Many کا تعلق براہ راست نہیں بن سکتا۔ ہمیں اس کے درمیان ایک تیسرا ٹیبل بنانا ہوگا۔

We create a new entity called: **BorrowRecord**

Instead of linking Member and Book directly, each borrowing activity is recorded as a separate entry in the BorrowRecord entity. This breaks the M:N relationship into two 1:M relationships.

BorrowRecord Entity Attributes:

- **BorrowID** (Primary Key): Unique ID for this specific transaction.
- **MemberID** (Foreign Key): Links to the Member who borrowed it.
- **BookID** (Foreign Key): Links to the specific Book being borrowed.
- *IssueDate* (Standard Attribute)
- *ReturnDate* (Standard Attribute)

Library Case Study (Final ER Schema)

Step 7: Final ER Diagram

The final structure shows:

1. One Member → Many BorrowRecords (1:M)
2. One Book → Many BorrowRecords (1:M)



4.8 Different Database Objects

In a DBMS, data is managed and presented using different tools known as database objects. These objects help users interact with the database efficiently. The four main objects are Tables, Forms, Queries, and Reports.

1. Tables

The most important object; it is where data is actually stored. A table is organized into rows (records) and columns (fields). Tables are the foundation of a database because all other objects depend on them.

یہ سب سے اہم حصہ ہے جہاں اصل ڈیٹا محفوظ ہوتا ہے۔

2. Forms

A Form provides a user-friendly interface for entering and updating data. Instead of typing directly into scary-looking tables, users use clean forms (like a web registration page) with text boxes, drop-downs, and buttons. Forms reduce data entry errors.

یہ ڈیٹا داخل (Enter) کرنے کا ایک آسان طریقہ ہے۔ اس سے غلطیوں کے امکانات کم ہوتے ہیں۔

Database Objects (Continued)

3. Queries

★ KEY CONCEPT

A Query is a request to retrieve specific data from the database. Queries allow you to "ask questions" of your database. For example: "Show me all students who scored above 80%." Queries are incredibly powerful because they filter, sort, and calculate data.

کیوری (Query) کا مطلب ہے ڈیٹابیس سے سوال پوچھنا، یعنی مخصوص معلومات نکالنا (جیسے صرف ان طلباء کی لسٹ جن کے نمبر 80 فیصد سے زیادہ ہیں)۔

4. Reports

A Report is used to present data in a structured, formatted, and visually appealing way, usually for printing or sharing. For example, a school generates a printed report card or a fee summary. You do not enter data into a report; you only read it.

یہ ڈیٹا کو خوبصورت اور منظم انداز میں پرنٹ کرنے یا پیش کرنے کے لیے استعمال ہوتا ہے (جیسے رزلٹ کارڈ)۔

💡 Quick Summary for Students

- **Tables:** Store the data.
- **Forms:** Enter the data.
- **Queries:** Search the data.
- **Reports:** Print the data.

4.9 Creation of Tables in MS Access

Creating tables is the fundamental step in developing a database. A carefully designed table ensures data accuracy, improves query speed, and saves storage.

4.9.1 Planning the Table Structure

Before touching the software, you must plan:

- **Purpose:** What will this table store?
 - **Fields:** What exact columns do I need?
 - **Data Types:** Is it Text, a Number, or a Date?
 - **Primary Key:** Which field is the unique identifier?
-

4.9.2 Creating Table using Datasheet View

Datasheet View is the fast, easy, spreadsheet-like way to create tables. It is great for beginners but offers limited control.

1. Go to the **Create** tab and select **Table**.
2. A blank grid appears (like Excel).
3. Double-click the column headers (Field1, Field2) to rename them.
4. Start typing data directly into the cells. MS Access tries to guess the data type automatically.

Limitations of Datasheet View

While fast, Datasheet view doesn't let you set strict validation rules (like "Marks cannot be over 100") easily. Professional developers prefer Design View.

Table Creation (Design View)

4.9.3 Creating Table using Design View

Design View provides full, structured control over your table before you enter any data. This is the professional way to build tables.

1. Go to the **Create** tab and select **Table Design**.
2. Enter the **Field Name** (e.g., StudentID, Name).
3. Select the exact **Data Type** from the dropdown.
4. Right-click the field that should be unique and select **Primary Key** (a small key icon will appear).
5. Save the table.

4.9.4 Selecting Appropriate Data Types

Choosing the right data type prevents errors and saves memory space.

Data Type	Used For
Short Text	Names, addresses, phone numbers (even though phone numbers look like math, you don't add or subtract them). Up to 255 chars.
Number	Numeric values used in mathematical calculations (e.g., Age, Marks, Quantity).
Date/Time	Dates of birth, admission dates, appointment times.
Currency	Financial values like Fees or Salary.
AutoNumber	Automatically generates a new unique sequential number for every new record. Perfect for Primary Keys.

4.10 Designing Forms

Forms serve as an interface to enter, modify, and view records. They minimize input errors and enforce validation rules visually.

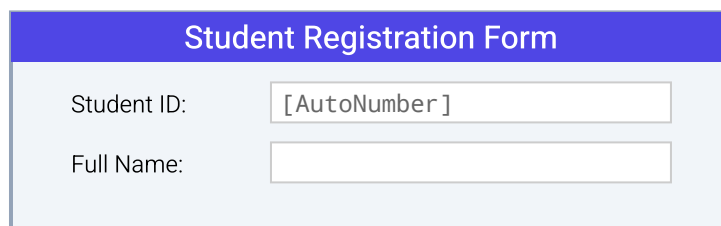
4.10.2 Identifying the Data Source

Every form **MUST** be linked to a data source (a Table or a Query). A form cannot exist without knowing where to save or retrieve its data.

4.10.3 Designing forms using Form Wizard

The Form Wizard provides a guided, step-by-step approach to creating clean forms.

1. Go to the **Create** tab and select **Form Wizard**.
2. **Select Table:** Choose your data source (e.g., Table: Student) from the dropdown.
3. **Select Fields:** Use the > button to move fields you want on the form, or >> for all fields.
4. **Choose Layout:** Select how the form looks. *Columnar* shows one record at a time and is best for data entry.
5. **Name & Finish:** Give the form a title (e.g., "Student Registration") and click Finish.



Student Registration Form	
Student ID:	[AutoNumber]
Full Name:	

4.11 Creating Queries in MS Access

★ GOLDEN TOPIC

Queries are the most powerful feature of Access. They retrieve, filter, and analyze data without changing the original tables.

4.11.2 Creating a Simple SELECT Query

A SELECT query simply picks and shows data. Steps to create one:

1. Go to the **Create** tab and click **Query Design**.
2. The "Show Table" dialog box appears. Select the table you want (e.g., Student) and click **Add**.
3. Double-click the fields you want to view (e.g., Name, Department). They will drop down into the design grid at the bottom.
4. Click the red **Run (!)** button in the ribbon to execute the query.

Field:	StudentName	Department	
Table:	Student	Student	
Criteria:			

Filtering Query Data

4.11.3 Filtering Data

By default, a query shows ALL rows. To see only specific records (e.g., only students in "Computer Science"), we type rules into the **Criteria** row of the design grid.

Single Criteria

Type "Computer Science" under the Department column. Only those students will appear.

Multiple Criteria (Logical AND)

If you want students who are named "Ali" **AND** are in "Computer Science", you put both criteria on the **SAME row** in the grid.

	Field: Name	Field: Department
Criteria:	"Ali"	"Computer Science"
or:		

Multiple Criteria (Logical OR)

If you want students who are in "IT" **OR** "Computer Science", you put the criteria on **DIFFERENT rows** (using the "or:" row).

	Field: Department
Criteria:	"IT"
or:	"Computer Science"

Data Summarization & Statistics

4.11.4 Data Summarization and Grouping

Grouping allows you to combine records with similar values (e.g., Grouping by Department) to calculate totals. In MS Access, click the **Totals (Σ)** button in the ribbon to enable the "Total" row in the grid.

Aggregate Function	What it calculates
COUNT()	Counts the number of records (e.g., How many students in IT?)
SUM()	Adds total values together (e.g., Total fees collected)
AVG()	Calculates the mathematical mean (average)
MAX() / MIN()	Finds the highest / lowest value in a group

4.11.5 Statistical Analysis Example

To find the Average Marks of students in each Department:

1. Create Query, add Student table.
2. Add **Department** and **Marks** fields to the grid.
3. Click the **Totals (Σ)** button.
4. Under Department, set Total to **Group By**.
5. Under Marks, set Total to **Avg**.
6. Run!

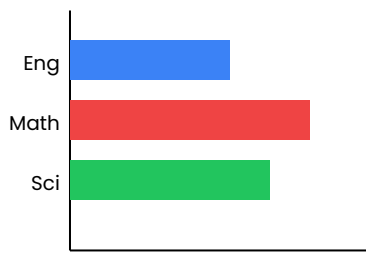
Department	Avg of Marks
Computer Science	82.5
IT	76.0

4.11.6 Data Visualization (Charts)

Data visualization transforms raw numbers into graphical representations, making patterns and trends instantly obvious.

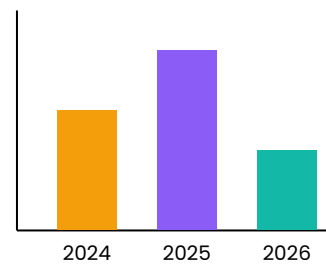
1. Bar Chart

Uses horizontal bars. Great when category names are long.



2. Column Chart

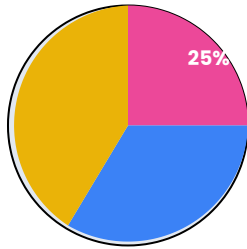
Uses vertical bars. Excellent for comparing categories side-by-side.



Data Visualization (Continued)

3. Pie Chart

Divides a circle into slices to show percentages (parts of a whole). Best for showing proportions.



4. Line Chart

Connects data points with lines. The absolute best choice for showing trends over time (e.g., website visitors per month).

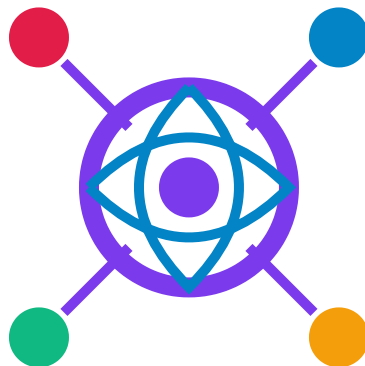


Summary

- **Data vs Info:** Data is raw facts; Information is processed data.
- **DBMS:** Software to create and manage databases securely and efficiently.
- **Table Components:** Rows are Records; Columns are Fields.
- **Keys:** Primary Key is unique for each record. Foreign Key links tables together.
- **ER Model:** Conceptual blueprint using Entities (rectangles), Attributes (ovals), Relationships (diamonds).
- **Many-to-Many:** Requires a Junction Table to resolve in relational databases.
- **Referential Integrity:** Ensures relationships stay valid via Cascade Update/Delete.
- **Access Objects:** Tables (store), Forms (enter), Queries (search/filter), Reports (print).
- **Data Vis:** Bar/Column for category comparison, Pie for parts of a whole, Line for trends.

COMPUTER SCIENCE XI

CHAPTER 5: APPLICATION AND IMPACTS OF COMPUTING



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

یہ نوٹس خاص طور پر اساتذہ کے لیے تیار کیے گئے ہیں۔ حوالہ کے لیے نصابی کتاب کی پی ڈی ایف فائل "CS XI (5 & 6).pdf" کا استعمال کریں، جس کی من و عن پیروی کی گئی ہے۔ جن موضوعات پر ★ کا نشان ہے وہ امتحانات کے لیے انتہائی اہم ہیں۔

These notes are prepared specifically for teachers. For reference, please consult the textbook PDF file named exactly "**CS XI (5 & 6).pdf**", which has been followed verbatim. Topics marked with a ★ are highly critical for examinations.

5.1 Introduction to Computing

Computing is now an important part of our daily lives. We use computers, mobile phones, the internet, and smart devices everywhere. In Pakistan, it is becoming vital in schools, hospitals, banks, farms, and industries.

کمپیوٹنگ اب ہماری روزمرہ زندگی کا ایک اہم حصہ ہے۔ پاکستان میں سکولوں، ہسپتالوں، بینکوں، کھیتوں اور صنعتوں میں اس کا استعمال تیزی سے بڑھ رہا ہے۔

5.1.1 Artificial Intelligence (AI)

★ GOLDEN TOPIC

AI is called "artificial" because it is created by humans, and "intelligence" means it can perform tasks that require thinking. AI systems use data to make decisions.

مصنوعی ذہانت (AI) انسانوں کی بنائی ہوئی ایک ایسی ٹیکنالوجی ہے جو سوچنے اور سمجھنے والے کام کر سکتی ہے۔ یہ فیصلے کرنے کے لیے ڈیٹا کا استعمال کرتی ہے۔

- *Example 1:* Mobile phone face recognition to unlock screens.
- *Example 2:* YouTube suggesting videos based on your interests.
- *Example 3:* Chatbots answering questions on a website.

AI in Smart Systems

AI often acts as the "**brain**" of a system, working together with the Internet of Things (IoT) and Data Analytics. IoT devices gather data, Analytics sorts it, and AI makes the final decision.

اسمارٹ سسٹمز میں AI ایک "دماغ" کے طور پر کام کرتا ہے۔ IoT آلات ڈیٹا جمع کرتے ہیں، اینالٹکس اسے ترتیب دیتا ہے، اور AI حتمی فیصلہ کرتا ہے۔

(e.g., If a room is hot, AI decides to turn on the AC automatically based on sensor data).



5.1.2 Internet of Things (IoT)

★ GOLDEN TOPIC

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خود کار طریقے سے ڈیٹا کا تبادلہ کرتے ہیں۔

Example: A smartwatch collects heart rate data and sends it to a mobile phone to display.

Components of IoT Systems

IoT relies on six foundational components to work properly:

1. Sensors (سینسرز)

Devices that collect information from the physical world. They are the "eyes and ears" of the system.

یہ آلات ارد گرد کے ماحول سے معلومات (جیسے درجہ حرارت، حرکت، دھواں) جمع کرتے ہیں۔

- Temperature sensor
- Motion sensor
- Fingerprint sensor
- Smoke / Soil moisture sensor

2. Actuators (ایکچویٹرز)

Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.

یہ آلات سسٹم کی ہدایات پر عمل کرتے ہوئے کوئی جسمانی کام (جیسے موٹر چلانا، دروازہ کھولنا) انجام دیتے ہیں۔

- Electric Motor / Switch
- Automatic door opener
- Water valve / Fan controller

IoT Components (Continued)

3. Connectivity

The connection that allows devices to send and receive data. Without it, sensors can't talk to the system.

Examples: Wi-Fi, Bluetooth, ZigBee, 4G/5G.

4. Data Processing

Changing raw data into useful info. Steps include collecting, organizing, analyzing, and deciding.

Example: Checking if room temp is too hot to trigger the AC.

5. Cloud Computing

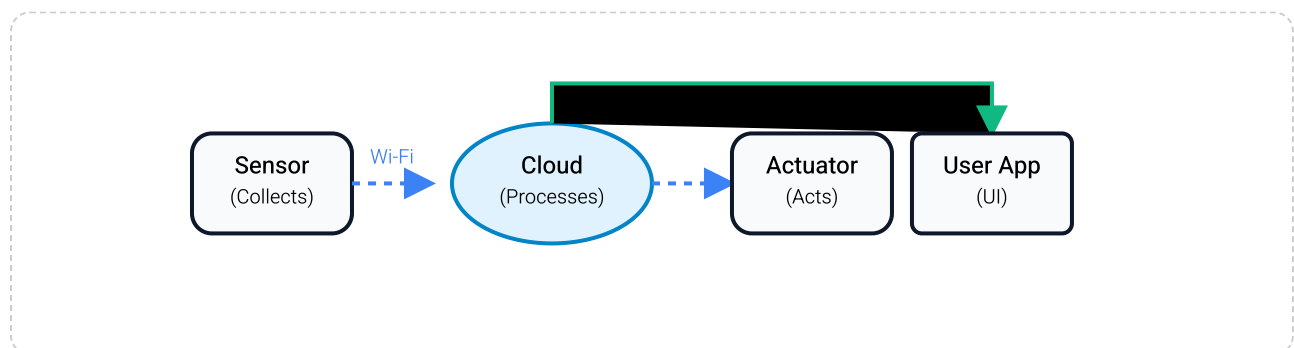
Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.

Examples: Google Drive, iCloud, Online LMS.

6. User Interface (UI)

The component that enables interaction between the user and the device. Helps users see info and give commands.

Examples: Mobile Apps, Touch screens, Voice commands.



5.1.3 Data Analytics

Data Analytics is the process of collecting, organizing, cleaning, studying, and presenting data to find useful information. It helps people make better decisions.

ڈیٹا اینالٹکس (Data Analytics) معلومات کو جمع کرنے، منظم کرنے، صاف کرنے اور ان کا تجزیہ کرنے کا عمل ہے تاکہ بہتر فیصلے کیے جاسکیں۔

Example: A college collects exam marks, calculates percentages, and identifies which students need extra help. This entire process is Data Analytics.

Components of Data Analytics

It comprises five key components:

1. Data Collection

Gathering raw information from multiple sources (websites, surveys, sensors, hospital records, bank transactions).

2. Data Storage

Storing data safely and properly after collection. It can be stored in files, databases, spreadsheets, or the cloud.

3. Data Cleaning

Identifying and correcting errors in data. Raw data may contain:

- Duplication of records
- Incorrect values (typos)
- Missing files or records
- Improper formats

Data Analytics (Continued)

4. Data Analysis

Studying the clean data to identify patterns, trends, and relevant information for problem-solving. (e.g., Finding out which subjects students failed the most).

5. Data Visualization

Creating visual representations of data (plots, charts, graphs) to provide valuable insights at a glance. (e.g., A pie chart showing pass vs fail percentage).

Comparison: IoT vs AI vs Data Analytics

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to internet.	Technology enabling machines to think & decide.	Process of examining data to find patterns.
Main Purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of Work	Works with physical devices (Hardware).	Works with smart software systems.	Works with data (Numbers, records).
Output	Raw data from environment.	Smart decisions or actions.	Reports, graphs, insights.

5.2 Uses of IoT in Different Areas

IoT improves efficiency, saves time, and enhances the overall quality of life. It plays a significant role in developing modern infrastructure.

آئی او ٹی (IoT) کارکردگی کو بہتر بناتا ہے، وقت بچاتا ہے، اور جدید انفراسٹرکچر (جیسے اسمارٹ سٹیز) بنانے میں اہم کردار ادا کرتا ہے۔

1. Smart Cities

- **Traffic Management:** Sensors adjust traffic signals automatically to reduce congestion.
- **Street Lighting:** Lights turn ON/OFF based on daylight or movement.
- **Parking:** Sensors guide drivers to empty parking spaces.
- **Waste Management:** Smart bins send alerts when full.

2. Industrial Automation

- Sensors check machine temperature and vibration to prevent breakdowns.
- Automated machines perform repeated tasks precisely.
- Quality control sensors check products during manufacturing (e.g., Textile industries in Sindh).

3. Healthcare (Smart Health)

- Smartwatches continuously measure heart rate and vitals.
- Remote monitoring devices send patient data directly to doctors (vital for rural areas in Pakistan).
- Emergency alert devices inform hospitals instantly.

Uses of IoT (Continued)

4. Education (Smart Learning)

- **Smart Classrooms:** Digital boards and connected projectors.
- **Attendance Systems:** Biometric automated attendance for students and teachers.
- **Resource Management:** Automated classroom lighting and AC control.

5. Home Automation

- **Smart Lighting:** Controlled remotely via mobile apps.
- **Security:** Real-time monitoring via Wi-Fi cameras.
- **Voice Control:** Using assistants (like Alexa/Google) to control TVs and appliances.

6. Smart Farming

Crucial for agriculture in Pakistan (especially Sindh/Punjab) to save water and improve yields.

- **Soil Moisture Sensors:** Check if crops actually need water.
- **Automated Irrigation:** Turns water pumps ON/OFF based on soil data.
- **Drones:** Monitor crop health from above.



5.2.1 AI in Education in Pakistan

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving teaching, learning, and administration.

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت (AI) کی اہمیت بڑھ رہی ہے، جس سے پڑھانے اور سیکھنے کے عمل میں بہتری آرہی ہے۔

Practical Uses

- Learning Management Systems (LMS).
- Automated grading and digital testing.
- AI-based career counselling.

Benefits

- Helps students learn at their own speed.
- Identifies students who need extra help.
- Supports students with disabilities (Assistive AI).

Major Challenges in Pakistan

- Limited internet access in rural areas.
- Lack of computers and digital devices for all students.
- Need for proper teacher training to use AI tools.
- Data privacy and security issues.

5.2.2 Career Connection in IoT and AI

Learning computing skills opens doors to software houses, telecom (Jazz, Telenor), banks, and government IT departments in Pakistan.

- **IoT Engineer:** Designs smart systems and connected devices.
- **Data Analyst:** Analyzes data generated by IoT systems.
- **Key Skills Needed:** Programming (Python, C++), Networking (Wi-Fi, protocols), Electronics (sensors/circuits), and Problem-Solving.

5.3 Information Sources

★ GOLDEN TOPIC

Information sources are ways to find knowledge and data. While the internet has vast information, not all of it is accurate. Sources are divided into three main categories:

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں۔ انہیں تین بڑی اقسام میں تقسیم کیا جاتا ہے: پرائمری، سیکنڈری، اور ٹرشری۔

1. Primary Sources (پرائمری ذرائع)

Give original, **firsthand information** straight from people, events, or official records. They are highly authentic.

Examples: Interviews, photographs, diaries, NADRA official records.

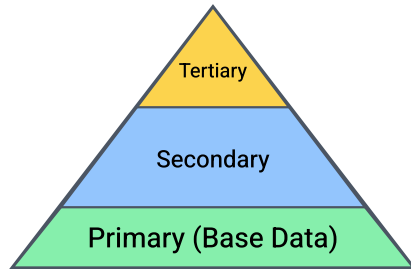
یہ براہ راست اور اصل معلومات فراہم کرتے ہیں (جیسے انٹرویوز، نادرا کا ریکارڈ)۔

2. Secondary Sources (سیکنڈری ذرائع)

Information that has **already been collected and explained by someone else**. They summarize or analyze primary data.

Examples: Textbooks, news reports, research papers, Bureau of Statistics reports.

یہ معلومات کسی اور کی جمع کردہ ہوتی ہیں اور پرائمری ڈیٹا کی وضاحت کرتی ہیں (جیسے نصابی کتابیں، نیوز رپورٹس)۔



Information Sources (Continued)

3. Tertiary Sources (ثالثی ذرائع)

Provide highly **summarized and organized information** derived from both primary and secondary sources. Used for quick reference.

Examples: Encyclopedias, Wikipedia, Dictionaries, Handbooks.

یہ پرائمری اور سیکنڈری ذرائع کا خلاصہ پیش کرتے ہیں، جو فوری حوالے کے لیے استعمال ہوتے ہیں (جیسے ویکیپیڈیا، ڈکشنری)۔

When do sources become unreliable?

In the digital age, students must check facts carefully. A source is unreliable if:

Source Type	Reasons for Unreliability
Primary Sources	Information is incomplete, data is recorded incorrectly, or false information was deliberately provided.
Secondary Sources	The information is outdated, lacks proper references/citations, or has not been verified by an expert.
Tertiary Sources	Comes from an unknown/unverified website, or the information is oversimplified and misses key facts.



Teacher's Tip

Teach students the rule of thumb: If an online article does not state the author's name, the date of publication, or list its references, it should not be trusted for academic work.

5.4 Impacts of Computing in Various Fields

Computing has drastically changed society. While it makes life faster and more connected, it also brings challenges like privacy risks and technology addiction.

Field	Positive Impact (+)	Negative Impact (-)
Education	Online classes, e-books, and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues.
Communication	Emails and video calls allow instant, global communication at zero cost.	People interact less in-person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking, and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Smart and Responsible Use

- Protect personal information and use strong passwords.
- Do not share fake news or unverified forwards.
- Balance screen time with physical sports and activities.

5.5 Assistive Technologies

★ GOLDEN TOPIC

Assistive technologies are devices, software, and systems that help people with disabilities or functional limitations. They allow more people to participate equally in school, work, and society.

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات اور سافٹ ویئر ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ تعلیم، کام اور معاشرے میں برابری کی بنیاد پر حصہ لے سکیں۔

Categories of Assistive Technologies:

1. Mobility Aids

Help people move from one place to another.

Examples: Smart wheelchairs, stair lifts.

2. Sensory Aids

Support vision and hearing to improve interaction.

Examples: Hearing aids, Screen readers (VoiceOver), Braille displays.

3. Communication Aids

Assist individuals with speech difficulties to express themselves.

Examples: Speech-generating devices, Text-to-speech apps.

4. Cognitive / Learning Aids

Help people with memory, reading, and understanding.

Examples: Audiobooks, spell-checkers, digital reminder apps.



Mobility



Sensory



Communication



Cognitive

Assistive Tech Importance & Careers

5.5.1 Importance of Assistive Technologies

- **Equal Access to Education:** Screen readers allow blind students to read digital textbooks equally with their peers.
- **Independence in Daily Life:** Mobility aids reduce the need for constant human assistance.
- **Social Inclusion:** Breaks down barriers, allowing individuals to work, study, and socialize confidently.

5.5.2 Career Connection

Designing assistive tech is a highly rewarding career path:

- **Assistive Software Developer:** Creates speech-to-text tools.
- **Biomedical Engineer:** Designs artificial limbs and hearing aids.
- **UX/UI Accessibility Expert:** Ensures websites are easy to navigate for visually impaired users.

Summary

- **AI:** Enables machines to learn, recognize patterns, and make decisions.
- **IoT Components:** Sensors (collect data), Actuators (take action), Connectivity, Data Processing, Cloud, and UI.
- **Data Analytics:** The pipeline of collecting, storing, cleaning, analyzing, and visualizing data for insights.
- **Information Sources:** Primary (direct/original), Secondary (analyzed by others), Tertiary (summarized lists/encyclopedias).
- **Impacts:** Computing accelerates business and education but risks privacy, job loss, and screen addiction.
- **Assistive Tech:** Tools like screen readers and smart wheelchairs that ensure equality, independence, and inclusion for people with disabilities.

COMPUTER SCIENCE XI

CHAPTER 6: DIGITAL LITERACY



BILINGUAL TEACHER'S EDITION

High-Yielding Lecture Notes & Golden Topics
Based on New Sindh Curriculum 2026

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Reference Note / اساتذہ کے لیے ہدایات

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6.1 Introduction to Digital Literacy

Digital literacy refers to the ability to understand and use digital devices, the internet, and online tools in a safe, effective, and responsible manner. In our increasingly digital world, possessing digital literacy is essential for everyone.

ڈیجیٹل لٹریسی کا مطلب ہے ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنے اور استعمال کرنے کی صلاحیت۔

It enables individuals to find information, communicate, produce digital content, and solve problems through technology.

Did you know?

Digital literacy is **not only using devices**; it is using digital information safely and wisely. A digitally literate student can locate information, organize it properly, turn it into something useful, and share it responsibly.

یہ صرف ڈیوائسز چلانا نہیں ہے، بلکہ معلومات کو محفوظ اور سمجھداری سے استعمال کرنا ہے۔

6.2 Digital Literacy in the Information Age

The Information Age is a period where vast amounts of data are produced, shared, and accessed instantly through the internet, mobile devices, and digital platforms.

انفارمیشن ایج ایک ایسا دور ہے جہاں بے تحاشا ڈیٹا پیدا، شیئر اور انٹرنیٹ کے ذریعے فوری حاصل کیا جاتا ہے۔

- Simply having access to technology is not enough.
- Students and users must develop the skills to understand, manage, and use information wisely.
- Digital literacy goes beyond basic computer skills; it involves technical, cognitive, and social abilities to function successfully in a digital society.

6.3 Types of Data

★ GOLDEN TOPIC

The data gathered through a survey, investigation, or classroom project, can be described as qualitatively and quantitatively. Understanding the difference is critical for proper analysis.

6.3.1 Qualitative Data

Describes what people think, feel, notice, or experience. It is non-numerical and descriptive.

کو الٹیٹیو ڈیٹا (Qualitative Data) یہ بتاتا ہے کہ لوگ کیا سوچتے ہیں، محسوس کرتے ہیں یا ان کا تجربہ کیا ہے۔ یہ غیر عددی (Non-numerical) ہوتا ہے۔

Examples: Students' opinions about online classes, descriptions of study habits, observation notes.

6.3.2 Quantitative Data

Deals with numbers. It is useful when students need to count, measure, compare, calculate averages, or display results in charts.

کو الٹیٹیو ڈیٹا (Quantitative Data) اعداد (Numbers) سے متعلق ہے۔ یہ گننے، ماپنے اور موازنہ کرنے کے کام آتا ہے۔

Examples: Number of students using the internet daily, marks obtained in a test, time spent online in hours.

6.3.3 Comparing Qualitative and Quantitative Data

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding <i>why</i> and <i>how</i> something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

6.4 Data–Collection Strategies

★ GOLDEN TOPIC

Digital literacy is not limited to finding information that already exists. Students should also learn how to gather original information. Below are the common strategies used to collect data.

ڈیجیٹل لٹریسی کا مطلب صرف موجودہ معلومات تلاش کرنا نہیں ہے، بلکہ اصل معلومات (Original Data) اکٹھا کرنے کے طریقے سیکھنا بھی ہے۔



6.4.1 Interviews

An interview is a planned conversation used to gather information directly from a person. It may be one-to-one or in a small group. Questions are planned in advance, and answers are usually qualitative.

یہ ایک طے شدہ گفتگو ہے جو کسی شخص سے براہ راست معلومات حاصل کرنے کے لیے کی جاتی ہے۔

6.4.2 Surveys / Questionnaires

A survey is useful when the same set of questions needs to be asked to many people. It may be on paper or created with digital tools like Google Forms or Microsoft Forms.

Note: Online survey tools automatically collect responses and display basic charts, making data analysis much easier.

Data-Collection Strategies (Continued)

6.4.3 Prototypes

A prototype is a simple early version of a product, model, or solution. It allows an idea to be tested before time and effort are spent on the final version. Users try the prototype and give feedback, which becomes useful data for improvement.

پروٹو ٹائپ کسی پروڈکٹ یا حل کا ابتدائی اور سادہ ورژن ہے۔ اس پر لوگوں کی رائے (Feedback) لی جاتی ہے۔

6.4.4 Observation

In observation, the researcher watches what people do or what happens in a situation and records the results carefully. No direct questions are asked. The data may be qualitative (notes) or quantitative (counts).

آبزر ویشن (Observation) میں محقق لوگوں کو دیکھتا ہے کہ وہ کیا کر رہے ہیں اور نتائج ریکارڈ کرتا ہے، بغیر کوئی سوال پوچھے۔

6.4.5 Simulations

A simulation is a computer-based model of a real situation. Variables can be changed to see what happens. Simulations are highly useful when real experiments are costly, risky, or difficult to perform.

یہ ایک کمپیوٹر پر مبنی ماڈل ہے۔ یہ اس وقت کارآمد ہے جب اصل تجربات مہنگے یا خطرناک ہوں۔

Teacher's Tip

Ask students which method they would use to find out "how many cars cross an intersection at 8 AM" vs "how students feel about the new cafeteria food". (Observation vs. Survey/Interview).

6.5 Primary and Secondary Data

★ GOLDEN TOPIC

Whenever data is used, it is important to know where it came from. The source affects how much control the researcher had over collecting it and how confidently the results can be used.

6.5.1 Primary Data

Primary data is collected **first-hand by the researcher** for a particular purpose. It is gathered directly from people, events, or situations rather than taken from an already published source.

پرائمری ڈیٹا وہ ہے جو محقق خود براہ راست کسی خاص مقصد کے لیے جمع کرتا ہے۔

Sources: Interviews, surveys, observations, experiments.

6.5.2 Secondary Data

Secondary data has **already been collected by someone else** and is being used again for a new purpose. It is extracted from existing archives.

سیکنڈری ڈیٹا وہ ہے جو پہلے سے کسی اور نے جمع کیا ہو اور اسے نئے مقصد کے لیے استعمال کیا جائے۔

Sources: Books, research articles, official reports, web statistics, online databases.

Primary Data

Fresh, Direct Collection
(You do the work)

Secondary Data

Existing Archives
(Someone else did it)

Primary vs Secondary & Approach Design

6.5.3 Difference Between Primary and Secondary Data

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

6.6 Designing a Data-Collection Approach

Before collecting data, a researcher must have a plan. This ensures accurate, relevant, and useful information is gathered.

1. **Define** the research question clearly.
2. **Decide** the type of data needed: quantitative, qualitative, or both.
3. **Select** suitable methods (survey, interview, observation).
4. **Choose** the sample or target group.
5. **Prepare** the tools such as forms, questions, and observation sheets.
6. **Collect** the data carefully and accurately.
7. **Store** the data digitally in a spreadsheet and keep a backup.
8. **Follow ethical practice** and respect privacy (Consent is mandatory).

6.7 Presenting Data Using Digital Tools

Collecting data is only one part of the work. Students must also present their findings so others can understand them quickly and easily. Digital tools make this much easier.

ڈیٹا جمع کرنا صرف ایک حصہ ہے۔ طلباء کو اپنے نتائج اس طرح پیش کرنے چاہئیں کہ دوسرے اسے آسانی سے سمجھ سکیں۔

6.7.1 Spreadsheets

Spreadsheets such as MS Excel, Google Sheets, and Apple Numbers help students arrange information in rows and columns.

- Enter data systematically.
- Calculate totals, averages, and percentages.
- Sort and filter records.
- Create charts and graphs automatically.

6.7.2 Presentations

Software like PowerPoint, Google Slides, and Canva helps students explain findings to an audience in an organized way.

- Begin with a clear title and research question.
- Use short bullet points instead of long paragraphs.
- Include tables, charts, and key findings.
- Keep the design simple and readable.



Presenting Data (Continued)

6.7.3 Infographics

An infographic is a visual way of presenting information, facts, or data so that people can understand the main idea quickly and easily. Tools such as Canva, Power BI, or Tableau can be used.

انفوگرافک معلومات اور حقائق کو بصری (Visual) انداز میں پیش کرنے کا طریقہ ہے، تاکہ لوگ اسے جلدی سمجھ سکیں۔

- Simplify complex concepts.
- Highlight the most important findings and results.
- Use clear visuals, limited text, and accurate information.

6.7.4 Reports

Reports are structured documents used to present collected data, findings, analysis, and conclusions. Tools like MS Word or Google Docs are standard.

- Start with a clear title, introduction, and purpose.
- Present the collected data using tables or charts.
- Explain findings in simple language.
- Include conclusions and recommendations based on the data.
- *Optional:* Use AI tools carefully to improve wording, but always review for accuracy.

Visual Rule of Thumb

A good chart or infographic can explain complex data faster than a long paragraph. However, visuals must always be accurate and based strictly on the real data collected, never exaggerated.

6.8 Case Study: Digital Inquiry Project

★ GOLDEN TOPIC

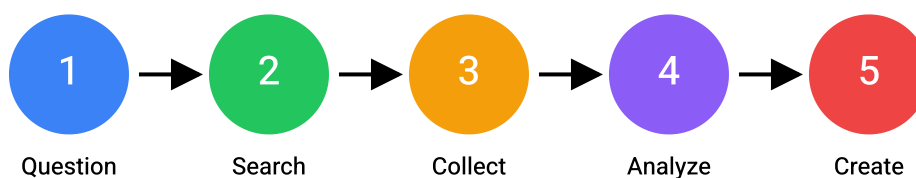
This comprehensive case study demonstrates how to apply all the concepts of Digital Literacy to solve a real-world problem from start to finish.

Research Question:

"How does daily mobile phone use affect the study habits of students?"

Scenario: Students notice classmates using phones during free periods and while studying. Some say it helps them learn; others say it wastes time. The class must investigate whether phone use helps or harms study habits.

The Digital Inquiry Workflow



Methodology (Steps 1 to 3)

Step 1: Perform Advanced Searches

Students first search for background information using search engines. They use **advanced search operators** to narrow down results and find reliable secondary data.

Suggested Advanced Search Queries	
"mobile phone use" "study habits" students	Uses quotes for exact phrases.
"academic performance" filetype:pdf	Finds only official PDF documents.
digital distraction site:.edu	Searches only educational websites.

Source Evaluation Checklist:

- Is the author/organization clearly mentioned?
- Is the information recent?
- Is the website reliable (.edu, .gov, research-based)?
- Does the source provide facts/references, or is it biased?

Step 2: Identify Data Types

Students must decide what kind of primary data they will collect.

- **Quantitative:** Number of hours used daily, percentage who feel distracted.
- **Qualitative:** Opinions on *why* they get distracted, suggestions for better use.

Step 3: Select Data-Collection Methods

Since they need to ask the same questions to many students, a **Survey/Questionnaire** (via Google Forms or paper) is the most efficient method.

Methodology (Steps 4 to 6)

Step 4: Prepare the Survey

Design clear, unbiased questions.

Question	Type of Data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (Categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Step 5: Choose the Sample

The **sample** is the group of people providing data. The students decide to survey 40 Class XI students from different sections to get a representative mix of habits.

Step 6: Collect Primary Data (Ethically)

When conducting the survey, students must follow strict **ethical guidelines**:

- Ask permission before collecting responses (Consent).
- Avoid collecting sensitive personal information.
- Do not record names unless absolutely necessary (Anonymity).
- Keep the responses private.
- Use the data *only* for educational purposes.

کسی کی ذاتی معلومات جمع کرنے سے پہلے اجازت لینا اور ان کی پرائیویسی کا خیال رکھنا ڈیجیٹل لٹریسی کا اہم حصہ ہے۔

Methodology (Steps 7 to 9)

Step 7: Use Secondary Data

Students include a reliable online article stating "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

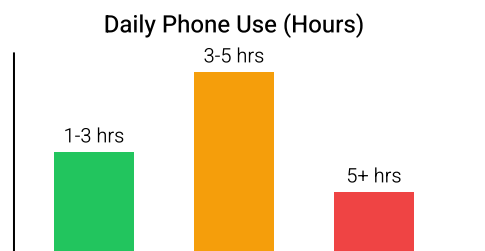
Step 8: Organize Data in a Spreadsheet

Responses are entered into MS Excel or Google Sheets to sort, calculate totals, and find averages.

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No

Step 9: Create Charts and Graphs

Convert the spreadsheet data into visuals to make it easy to digest.



Analysis and Conclusion (Steps 10–12)

Step 10: Analyze the Results

By looking at the spreadsheet and charts, the group identifies trends:

- Most students (majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: Students using phones mostly for entertainment report much higher distraction levels than those using them for study.

Step 11: Write the Conclusion

"The research shows mobile phones have both positive and negative effects. They provide access to learning materials. However, excessive use of social media and gaming reduces concentration. Therefore, phones should not be banned, but managed wisely."

Step 12: Recommendations

Based on the data, the students recommend:

- Turn off social media notifications while studying.
- Set daily screen-time limits.
- Take physical breaks from screens to protect health.

The Final Digital Artefact

A digital artefact is a final product created using digital technologies. Students must create one final artefact that answers the original research question. It is *not* just a decorated file; it must present evidence and conclusions.

یہ حتمی پروڈکٹ ہے جو ڈیجیٹل ٹیکنالوجی کے ذریعے بنائی جاتی ہے، جیسے رپورٹ یا پریزنٹیشن۔

Artefact Option	Suitable Digital Tools	Main Purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Standard Artefact Structure

1. Title
2. Research Question
3. Introduction
4. Method of Data Collection
5. Primary Data Findings
6. Secondary Data Summary
7. Charts and Graphs
8. Conclusion
9. Recommendations
10. References

Careers & Summary

Career Opportunities in Digital Research

- **Market Research Analyst:** Designs surveys to collect primary data to guide business decisions.
 - **UX (User Experience) Researcher:** Gathers qualitative/quantitative data to improve app and website designs.
 - **Data Journalist:** Uses advanced search skills to research stories and presents them via infographics.
 - **Digital Marketing Specialist:** Collects audience data and presents results through dashboards and reports.
-

Summary

- **Digital Literacy:** Not just using devices, but finding, evaluating, and creating information safely and responsibly.
- **Data Types:** Qualitative (opinions, words) vs. Quantitative (numbers, counts).
- **Data Sources:** Primary (collected first-hand by you) vs. Secondary (already collected by someone else).
- **Collection Methods:** Interviews, Surveys, Prototypes, Observation, and Simulations.
- **Advanced Search:** Using operators like quotes (" ") or filetype:pdf to find reliable sources efficiently.
- **Ethics:** Always ask for permission (consent) and respect user privacy when collecting data.
- **Digital Tools:** Spreadsheets (organize/calculate), Presentations (deliver to audience), Infographics (quick visual summaries), Reports (detailed structure).
- **Digital Artefact:** The final product that effectively communicates the answer to the research question using evidence and visuals.